



# SAMPLING BIAS IN HOUSEHOLD STUDIES

15 MAY 2024

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# OBJECTIVES

## Better understand the impact of age in the transmission dynamics in households

Development of a stochastic model  
of transmission **in households**<sup>1</sup>



Adaptation of the model in collaboration with M. Layan

**Data :** PedCovid longitudinal study  
PI : Pr. Isabelle Sermet-Gaudelus,  
Funding : PHRC + ANR



Statistical  
inference

Susceptibility

Infectiousness

Impact of age on  
transmission risk

Recruitment through identification of positive children: bias?

**Objective:** discuss how to correct/take into account this sampling bias

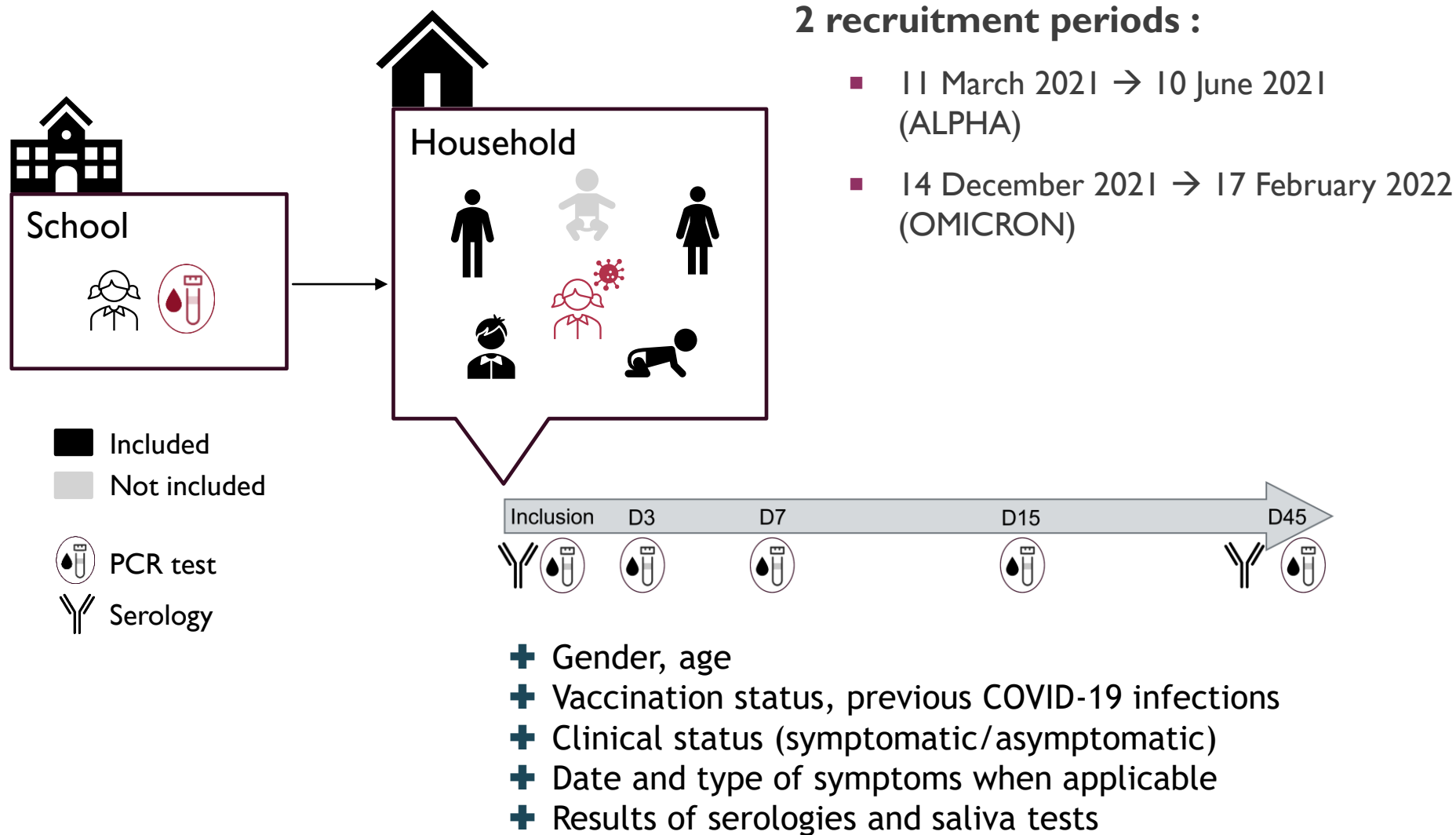
<sup>1</sup> Layan M, et al. Impact of BNT162b2 Vaccination and Isolation on SARS-CoV-2 Transmission in Israeli Households: An Observational Study. *Am J Epidemiol* 2022; **191**: 1224–34.



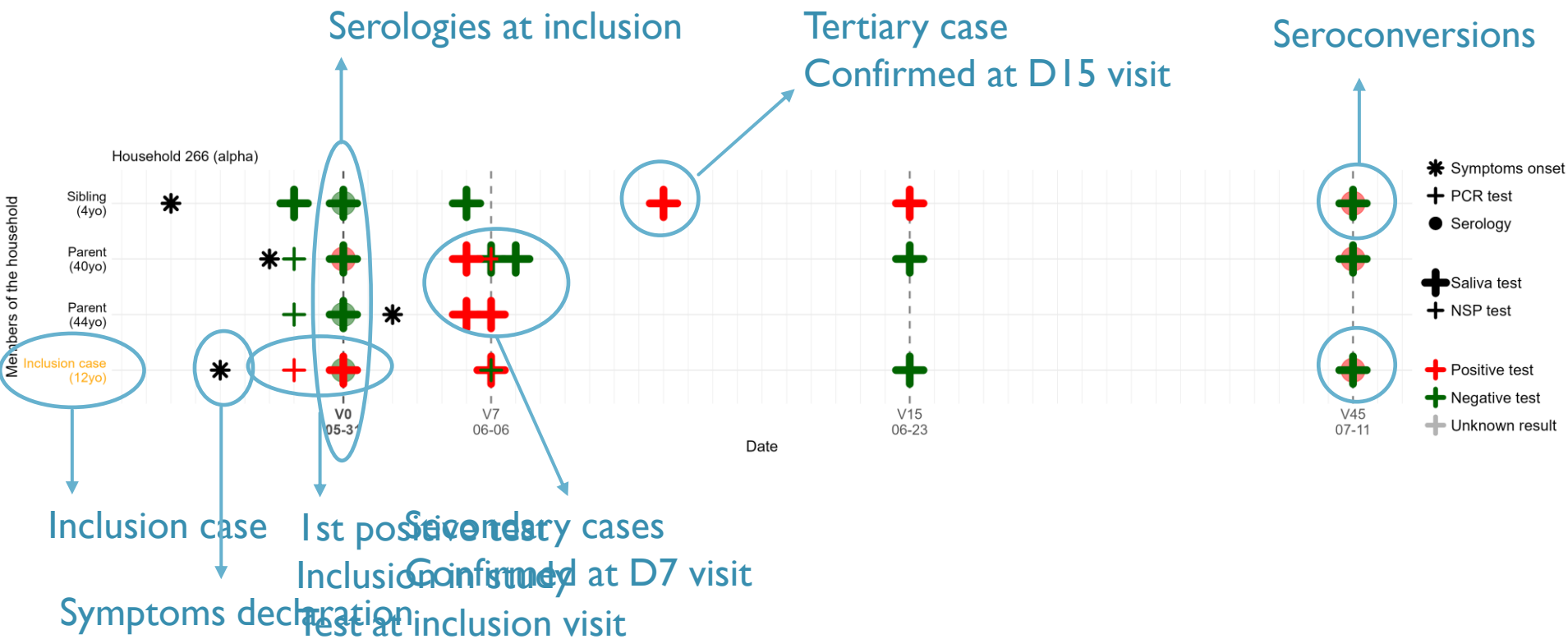
# I – DATA

DESIGN  
DESCRIPTION

# DESIGN



# HOUSEHOLD TRAJECTORIES



# HOUSEHOLD CHARACTERISTICS

|                               | Alpha period | Omicron period |
|-------------------------------|--------------|----------------|
| <b>N</b>                      | 128          | 54             |
| <b>Household size*</b>        | 4 (4-5)      | 4 (4-5)        |
| <b>N with AR=100%</b>         | 30 (23%)     | 23 (43%)       |
| <b>Age of inclusion case*</b> | 10 (7-13.25) | 6 (4.25-8)     |

\*median (IQR)



# 2 – METHODS

MODEL  
STATISTICAL APPROACH

# MODEL

- **Estimated parameters of interest:**

- Relative infectiousness of individuals across ages and symptoms statuses  $\rightarrow \mu_{inf}(s_j, a_j)$
- Relative susceptibility of individuals across ages  $\rightarrow \mu_{sus}(a_i)$

- **Age groups:**

- Infants (<6yo)
- Children (6-11 yo)
- Teens/adults (>11yo)

**Reference :** symptomatic infants (<6 yo)



# MODEL

## ■ Risk of infection of individual i by j at time t :

$$h_{j \rightarrow i}(t) = \underbrace{\frac{\beta}{(n_i/4)^\delta}}_{\text{Baseline transmission rate dependent on household size } (\delta)} \times \underbrace{\mu_{susc}(a_i) \times \mu_{acq}(p_i)}_{\text{Relative susceptibility of infected i : By age of i } (\mu_{susc}) \text{ By protection of i } (\mu_{acq})} \times \underbrace{\mu_{inf}(s_j, a_j) \times \mu_{transm}(p_j)}_{\text{Relative infectiousness of infector j : By age and symptomatic status of j } (\mu_{inf}) \text{ By protection of j } (\mu_{transm})} \times \underbrace{f(t - \tau_j)}_{\text{Likelihood of generation time}}$$

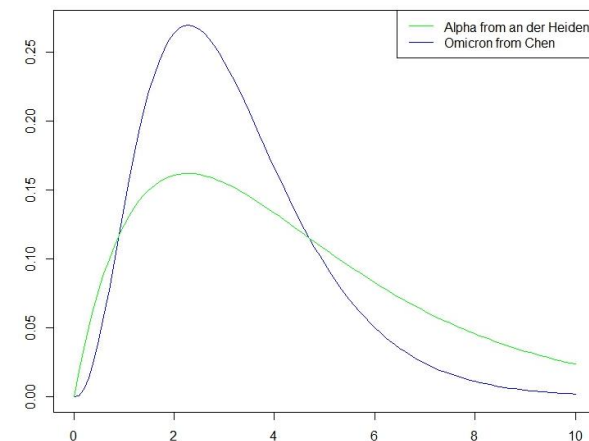
Baseline transmission rate  
dependent on household  
size ( $\delta$ )

Relative susceptibility of infected i :  
By age of i ( $\mu_{susc}$ )  
By protection of i ( $\mu_{acq}$ )

Relative infectiousness of infector j :  
By age and symptomatic status of j ( $\mu_{inf}$ )  
By protection of j ( $\mu_{transm}$ )

Likelihood of  
generation time

Distribution of generation time:



# MODEL

- Risk of infection of individual  $i$  at time  $t$  :

$$\lambda_i(t) = \underbrace{\alpha}_{\text{Risk of infection in the community}} + \underbrace{\sum_{j \in H_i, \tau_j < t} h_{j \rightarrow i}(t)}_{\text{Risk of infection in the household (individuals from the same household infected before } t)}$$

→ Likelihood

→ Markov Chain Monte Carlo to estimate parameters



## 3 – RESULTS

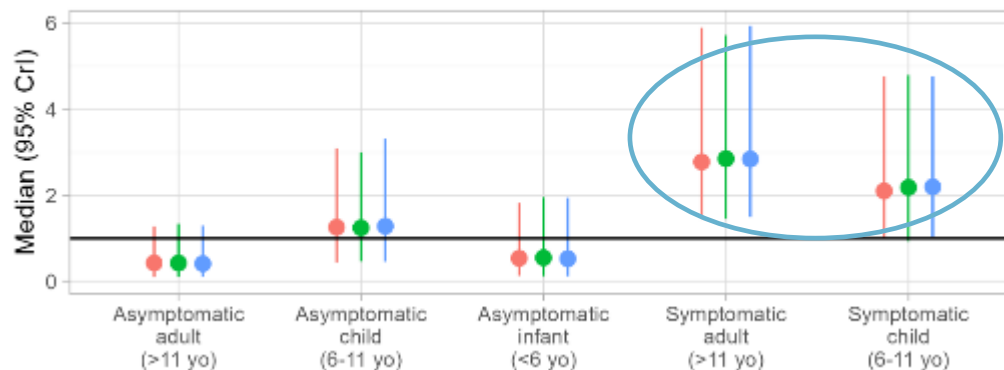
# RESULTS

## Relative infectiousness

alpha

A Relative infectivity

Reference: symptomatic infants (<6 yo)  $\beta=0.2$



Symptomatic infants (<6 yo) →

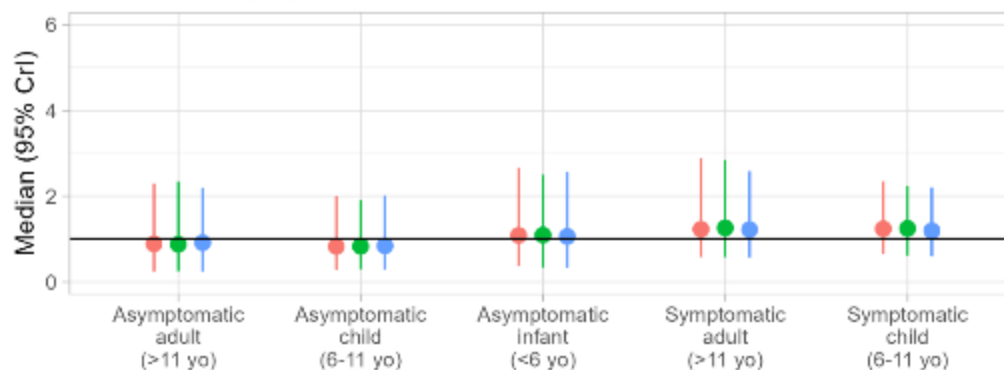
Alpha:

Among symptomatic individuals:  
Infants are less infectious than others

omicron

A Relative infectivity

Reference: symptomatic infants (<6 yo)  $\beta=0.632$



Symptomatic infants (<6 yo) →

Omicron:

No difference of infectiousness across ages and symptoms (lack of power?)

# RESULTS

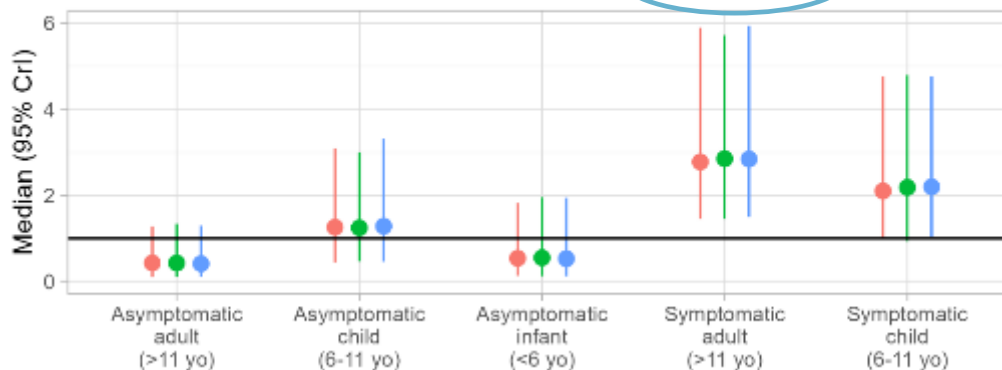
## Relative infectiousness

**alpha**

**A** Relative infectivity

Reference: symptomatic infants (<6 yo)

$\beta=0.2$



Symptomatic  
infants  
(<6 yo)

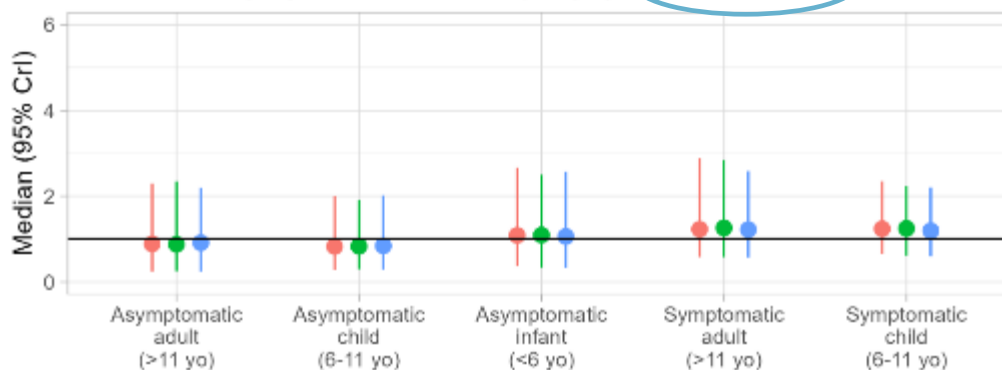
Baseline transmission rate  
higher for Omicron than  
for Alpha

**omicron**

**A** Relative infectivity

Reference: symptomatic infants (<6 yo)

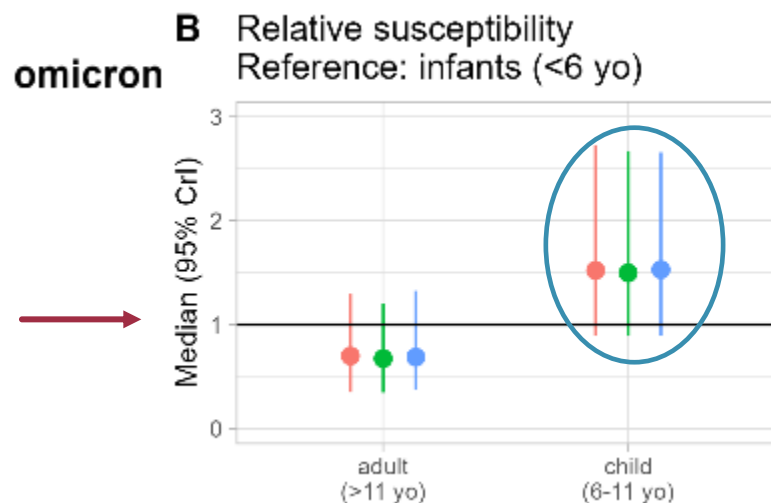
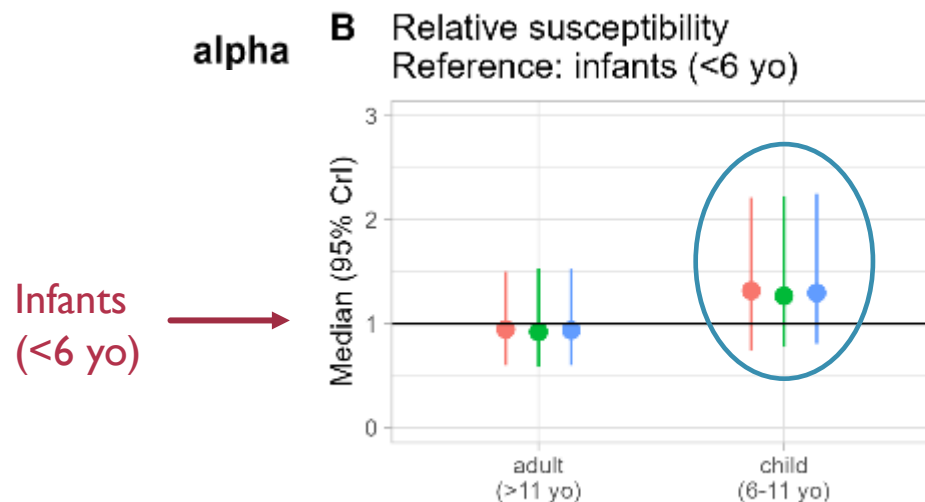
$\beta=0.632$



Symptomatic  
infants  
(<6 yo)

# RESULTS

## Relative susceptibility



### Tendency:

Children (6-11 yo) more susceptible than infants (<6yo)

# SUMMARY

## Summary :

- Convergence of the method ok
- Some statistically significant results

## Limits :

- Limited power (number of households / number of individuals in each age~symptoms category)
- Recruitment of households via children → bias?

→ Simulation study

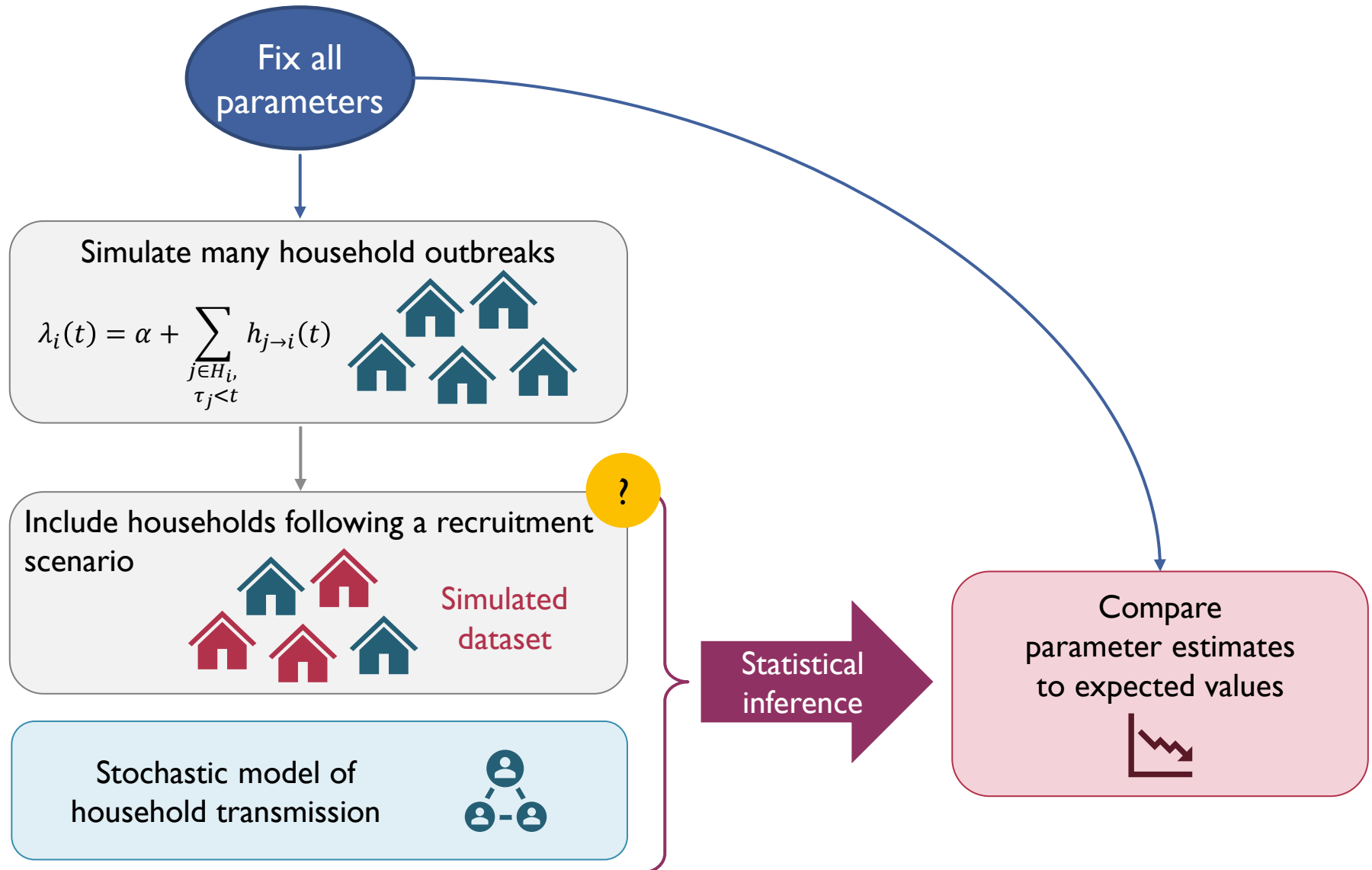


## 4 – SIMULATION STUDY



# SIMULATION STUDY

**Objective :** compare scenarios of households recruitment



# RECRUITMENT SCENARIOS

## Recruitment via **children** (PedCovid-like)



- If the inclusion case is an adult → exclude
- If the inclusion case is a child → recruit

## **Random** recruitment (Random)



- No classification/exclusion
- Draw randomly a total number of hh

## Recruitment via **adults** (Via adults)



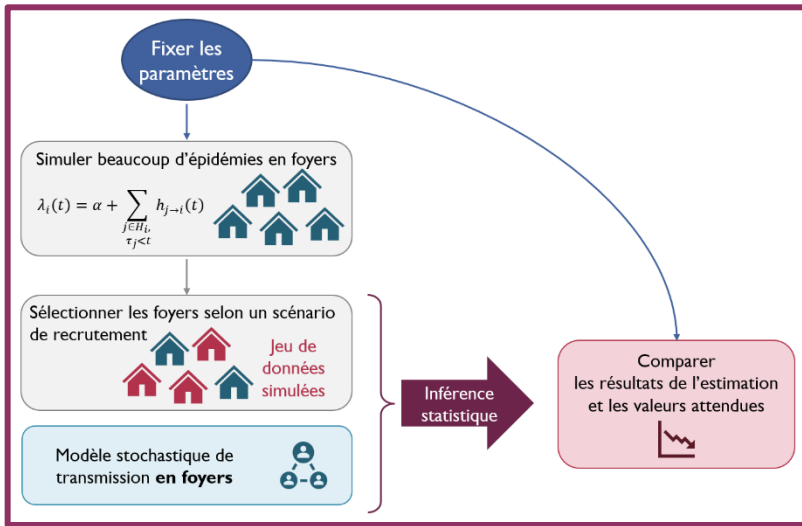
- If the inclusion case is a **child** → exclude
- If the inclusion case is an **adult** → recruit

## **Over-sampling** with recruitment via children (PedCovid-like + sampling)



- If the inclusion case is an adult → exclude
- **If no adult is infected in the household → exclude**
- If the inclusion case is a child → recruit

# SIMULATION STUDY



50 times

For each **recruitment scenario**

And each **parameter set**

## Tested parameter sets:

**Null hypothesis:** no difference in infectiousness/susceptibility across ages or symptoms

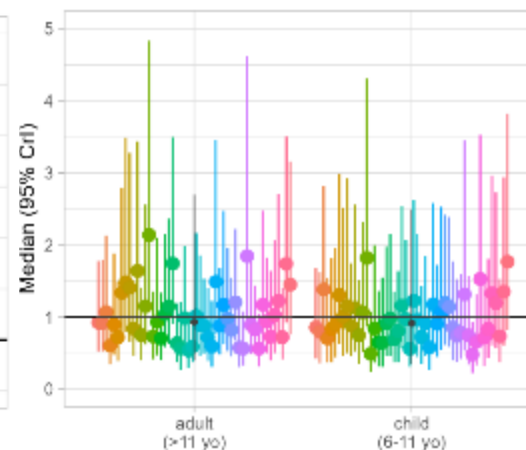
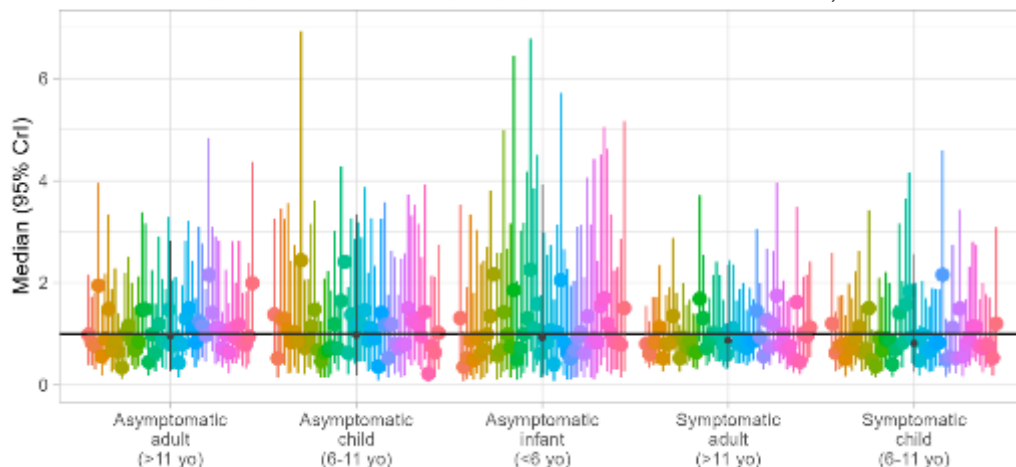
**Bias study:** vary parameter values around 1 independantly (while the others = 1)

# NULL HYPOTHESIS

Relative infectiousness = 1  
Relative susceptibility = 1

## Random recruitment (alpha)

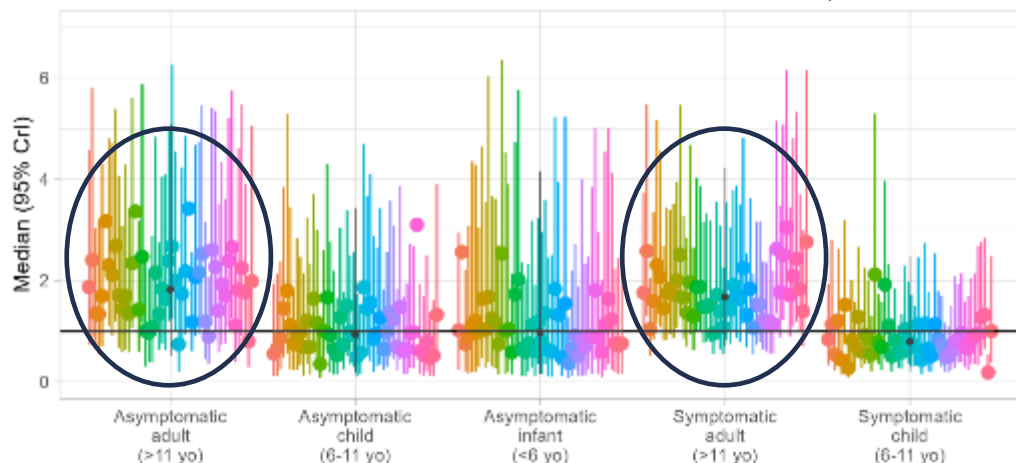
Relative infectivity  
Reference: symptomatic infants (<6 yo)  $\beta = 0.204$  (simu=0.2)



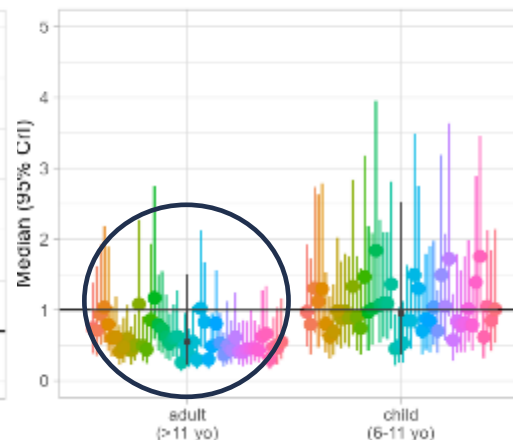
Random recruitment:  
Good estimation

## Recrutement PedCovid-like (alpha)

Relative infectivity  
Reference: symptomatic infants (<6 yo)  $\beta = 0.187$  (simu=0.2)



**B** Relative susceptibility  
Reference: infants (<6 yo)



PedCovid-like recruitment:  
Over-estimation of infectiousness in adults  
Under-estimation of susceptibility in adults

# NULL HYPOTHESIS

Relative infectiousness = 1  
Relative susceptibility = 1

→ Similar results for omicron variant

## Conclusion :

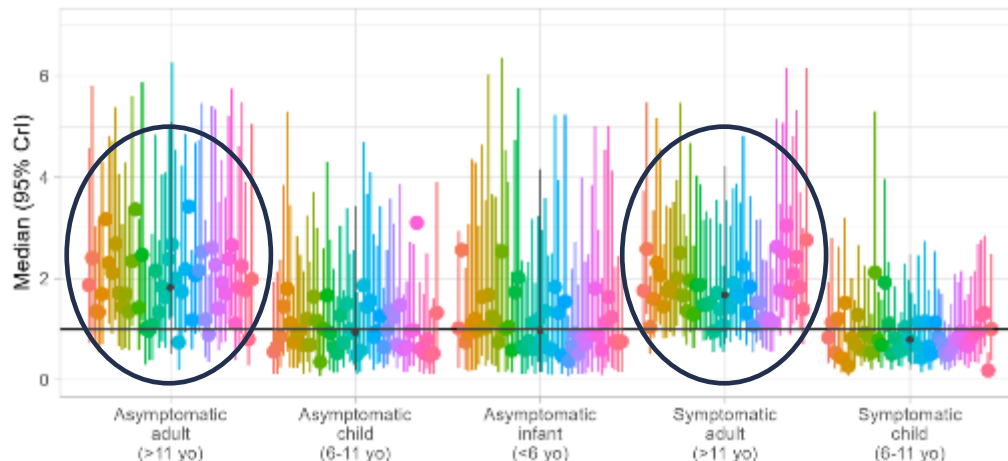
- Presence of a bias, potentially high and statistically significant
- Test values lower/higher than 1 to evaluate the impact of bias on our results

Random recruitment:  
Good estimation

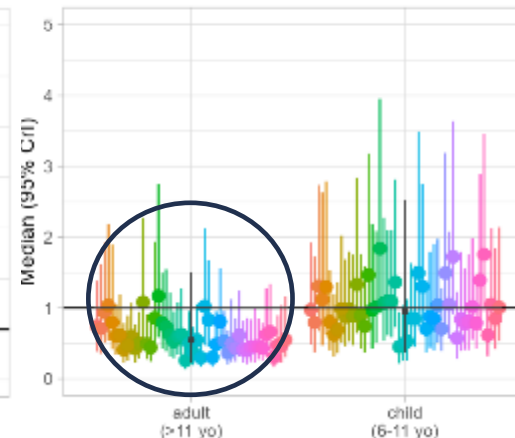
PedCovid-like recruitment:  
Over-estimation of infectiousness in adults  
Under-estimation of susceptibility in adults

### Recrutement PedCovid-like (alpha)

Relative infectivity  
Reference: symptomatic infants (<6 yo)  $\beta = 0.187$  (simu=0.2)



B Relative susceptibility  
Reference: infants (<6 yo)



# BIAS STUDY

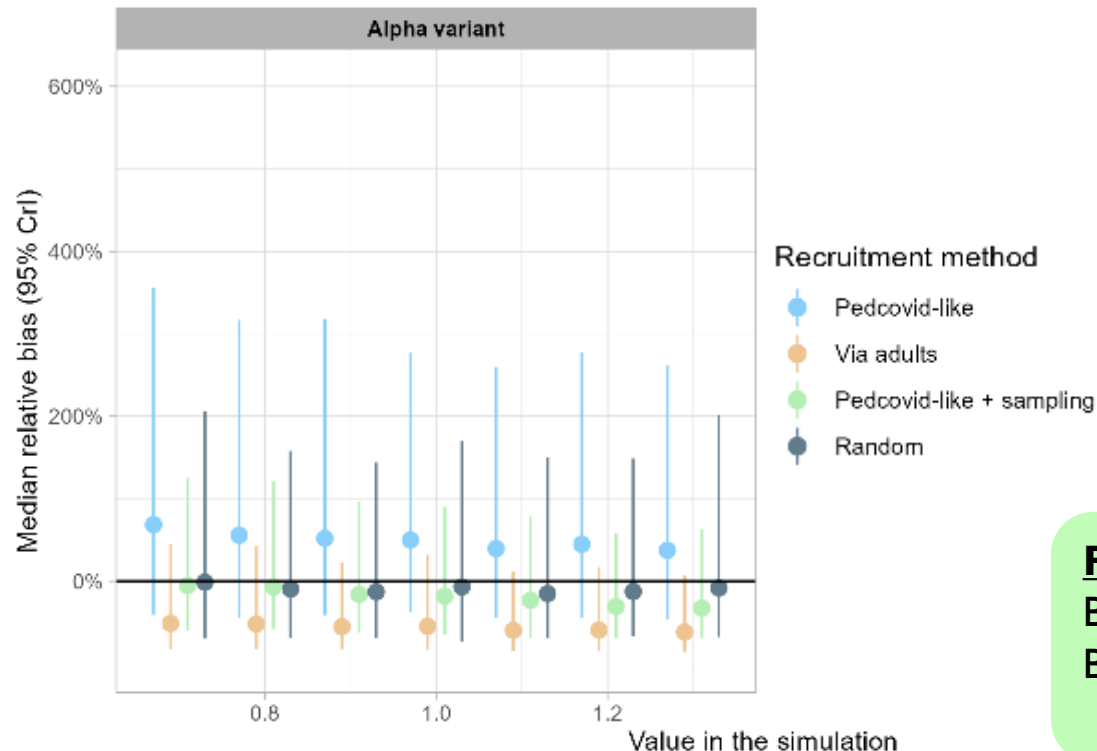
**Relative bias** =  $\frac{\text{estimated value} - \text{expected value}}{\text{expected value}}$  where the expected value is the one used in the simulations

Aggregated results of the 50 simulations for each recruitment scenario:

## Bias for Infectivity of Symptomatic Adults

50 aggregated results for each simulated value

Bias = (estimated value - simulation value) / simulation value



### PedCovid-like recruitment:

Over-estimation of infectiousness in symptomatic adults

### Random recruitment:

No apparent bias

### Recruitment via adults:

Under-estimation of infectiousness in symptomatic adults  
→ inverted bias as expected

### PedCovid-like recruitment + sampling:

Bias greatly reduced

But decreasing bias

→ probably not a good correction

# BIAS STUDY

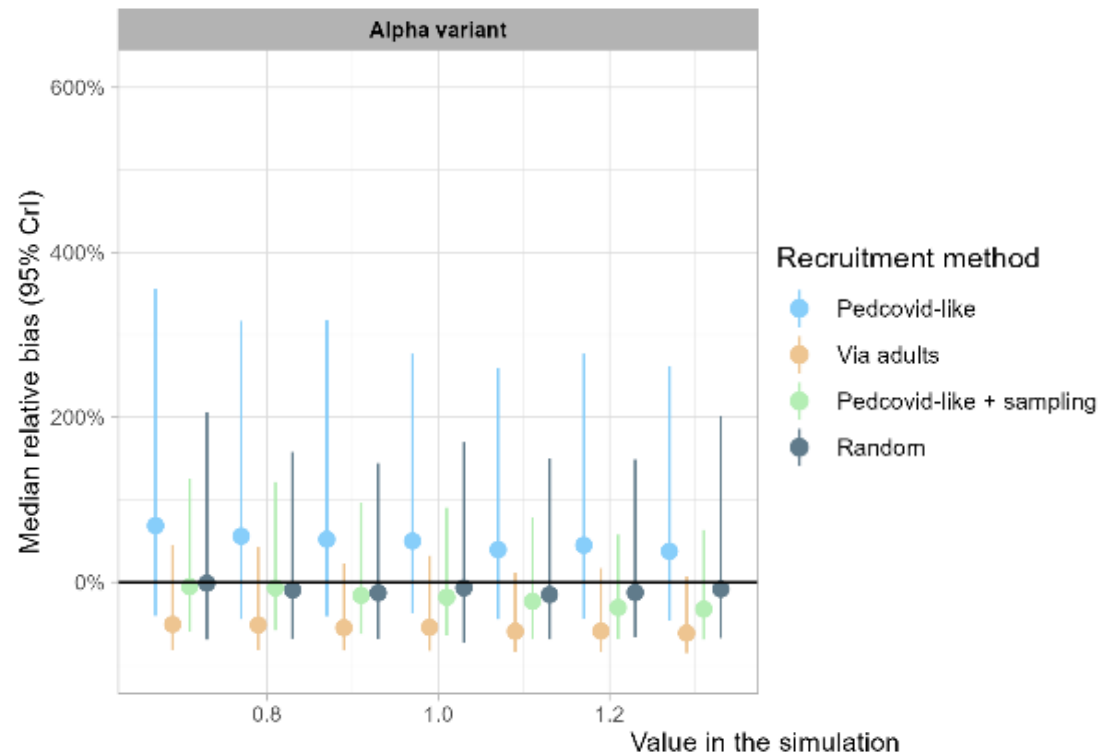
**Relative bias** =  $\frac{\text{estimated value} - \text{expected value}}{\text{expected value}}$  where the expected value is the one used in the simulations

Aggregated results of the 50 simulations for each recruitment scenario:

## Bias for Infectivity of Symptomatic Adults

50 aggregated results for each simulated value

Bias = (estimated value - simulation value) / simulation value



→ Similar results for omicron

→ Similar results for the other parameters

# CONCLUSION AND PERSPECTIVES

## Conclusion:

- Bias whatever the value of the parameter is
- Need to take it into account in the interpretation of results

## Perspectives for correction of bias:

- Other sampling method to explore (only hh where inclusion case is index?)
- Directly integrate the recruitment method in the likelihood computation (collaboration with Simon Cauchemez)
- Others?





# SUPPLÉMENTS



# DATA

# CARACTÉRISTIQUES DES INDIVIDUS

|                                        |              | Alpha period      |                      |                         | Omicron period    |                      |                         |
|----------------------------------------|--------------|-------------------|----------------------|-------------------------|-------------------|----------------------|-------------------------|
| N                                      |              | 535               |                      |                         | 234               |                      |                         |
| N of positive cases                    |              | 334 (62%)         |                      |                         | 179 (76%)         |                      |                         |
|                                        |              | Infants<br>(<6yo) | Children<br>(6-11yo) | Teens/adults<br>(>11yo) | Infants<br>(<6yo) | Children<br>(6-11yo) | Teens/adults<br>(>11yo) |
| N                                      |              | 61                | 124                  | 350                     | 60                | 49                   | 125                     |
| Sex F                                  |              | 31 (51%)          | 68 (55%)             | 187 (53%)               | 26 (43%)          | 17 (35%)             | 61 (49%)                |
| Age                                    |              | 4 (2-5)           | 9 (7-10)             | 38 (16-44)              | 3 (1-5)           | 8 (7-10)             | 39 (36-43)              |
| N inclusion cases                      |              | 21 (16%)          | 55 (43%)             | 52 (41%)                | 24 (44%)          | 24 (44%)             | 6 (11%)                 |
| Protected                              | Past infect° | 2                 | 1                    | 3                       | 3                 | 3                    | 2                       |
|                                        | Vaccinat°    | 0                 | 1                    | 27                      | 1                 | 2                    | 109(+2)                 |
|                                        | None         | 59                | 122                  | 320                     | 56                | 44                   | 14                      |
| Infection status                       | Not infect   | 23 (38%)          | 36 (29%)             | 142 (41%)               | 11 (18%)          | 5 (10%)              | 39 (31%)                |
|                                        | Sympto       | 28                | 61                   | 184                     | 39                | 35                   | 69                      |
|                                        | Asympto      | 10                | 27                   | 24                      | 10                | 9                    | 17                      |
| Chez les infectés, prévalence asympto: |              | 26%               | 31%                  | 12%                     | 20%               | 20%                  | 20%                     |

# CARACTÉRISTIQUES DES INDIVIDUS

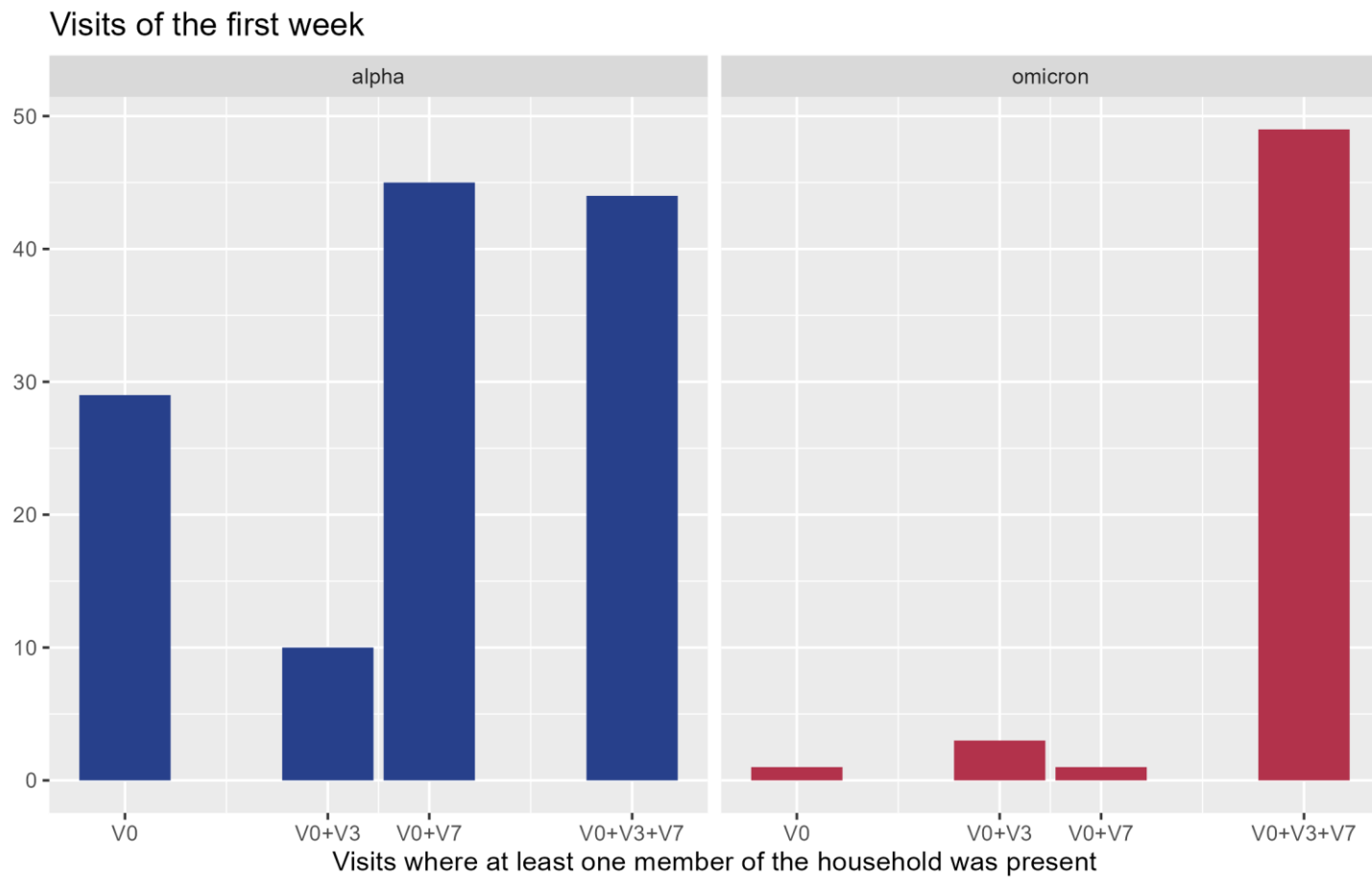
## Exclusion des cas index

### Alpha period

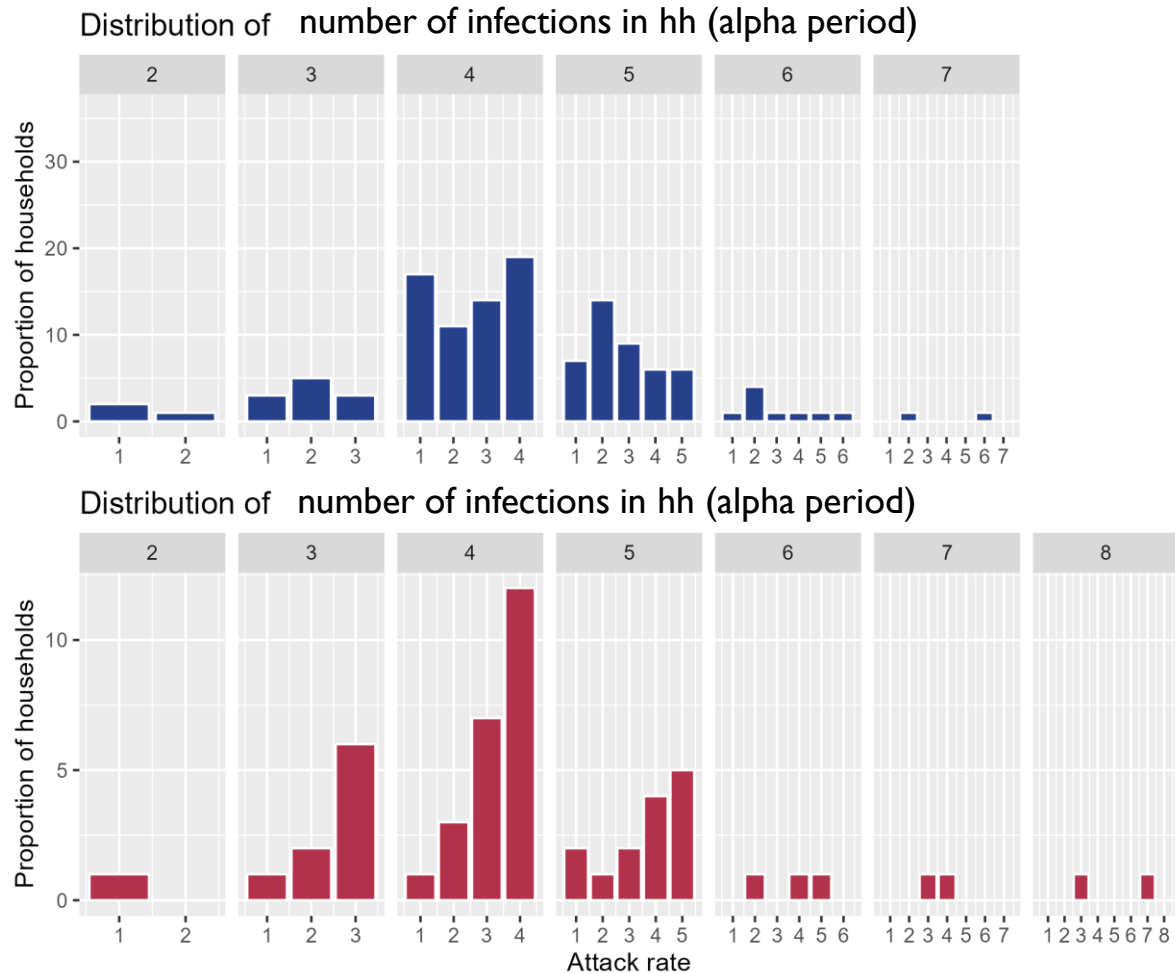
### Omicron period

| N                                      |              | 407            |                   |                      | 180            |                   |                      |
|----------------------------------------|--------------|----------------|-------------------|----------------------|----------------|-------------------|----------------------|
| N of positive cases                    |              | 206 (51%)      |                   |                      | 125 (69%)      |                   |                      |
|                                        |              | Infants (<6yo) | Children (6-11yo) | Teens/adults (>11yo) | Infants (<6yo) | Children (6-11yo) | Teens/adults (>11yo) |
| N inclusion cases                      |              | 21 (16%)       | 55 (43%)          | 52 (41%)             | 24 (44%)       | 24 (44%)          | 6 (11%)              |
| N                                      |              | 40             | 69                | 298                  | 36             | 25                | 119                  |
| Sex F                                  |              | 21 (53%)       | 42 (61%)          | 159 (53%)            | 16 (44%)       | 17 (44%)          | 61 (50%)             |
| Age                                    |              | 3 (1-5)        | 9 (7-10)          | 40 (34-45.75)        | 2 (1-3.25)     | 8 (7-10)          | 39 (36-43)           |
| Protected                              | Past infect° | 1              | 0                 | 3                    | 3              | 2                 | 2                    |
|                                        | Vaccinat°    | 0              | 0                 | 27                   | 0              | 1                 | 104(+2)              |
|                                        | None         | 39             | 69                | 268                  | 3              | 22                | 13                   |
| Infection status                       | Not infect   | 23 (58%)       | 36 (52%)          | 142 (48%)            | 11 (31%)       | 5 (20%)           | 39 (33%)             |
|                                        | Sympto       | 12             | 22                | 138                  | 19             | 16                | 63                   |
|                                        | Asympto      | 5              | 11                | 18                   | 6              | 4                 | 17                   |
| Chez les infectés, prévalence asympto: |              | 29%            | 33%               | 12%                  | 24%            | 20%               | 21%                  |

# CARACTÉRISTIQUES DES FOYERS

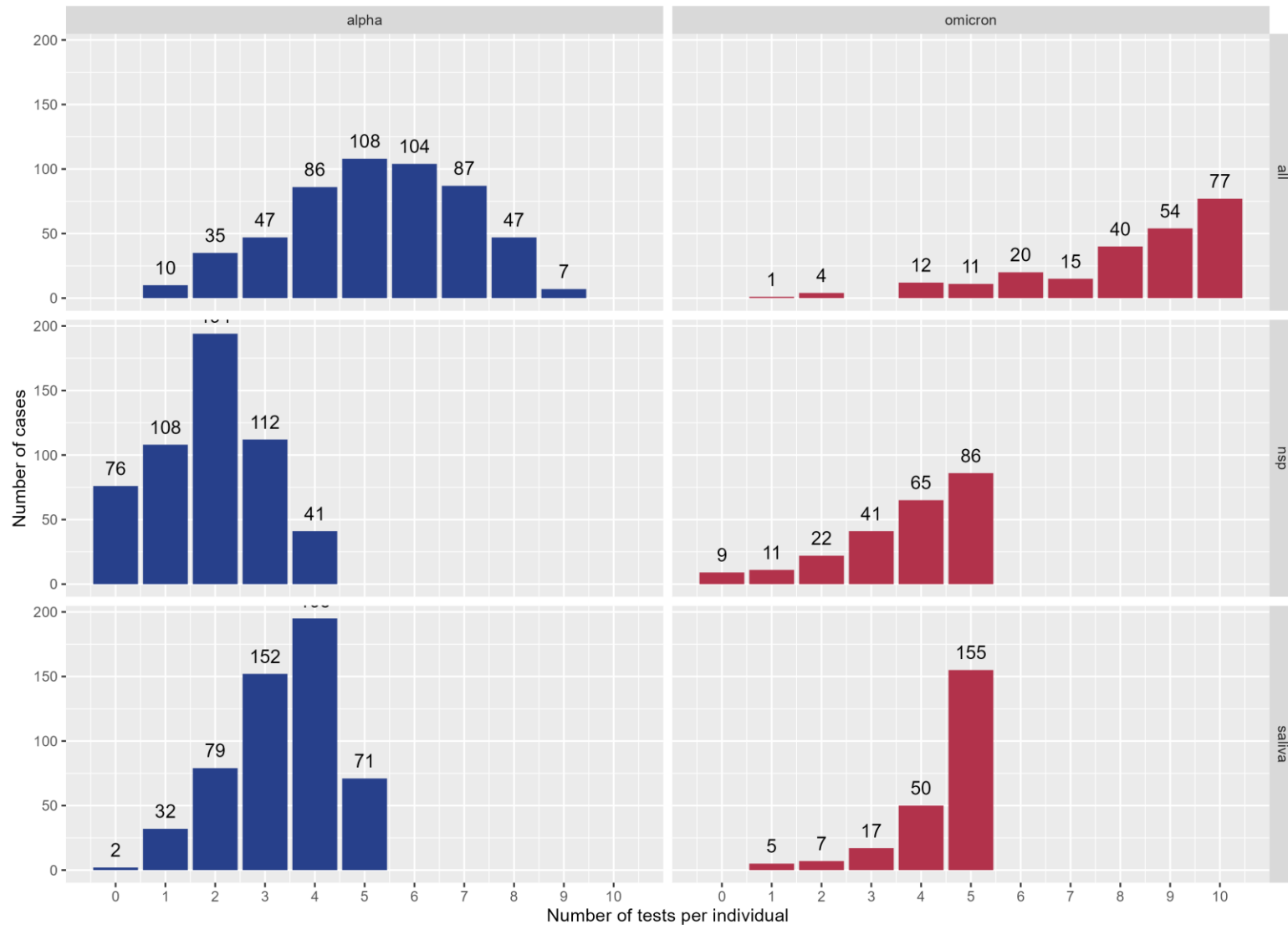


# ATTACK RATES



# CARACTÉRISTIQUES DES INDIVIDUS

Distribution of number of tests by individual (2295)





# MODEL



# HYPOTHESES

## ■ Data from individual $i$ of household $H_i$ :

- Individual id
- Household id ( $H_i$ )
- Household size ( $n_i$ )
- Symptoms onset date (for symptomatics – 1<sup>st</sup> **positive test** for asymptomatics) ( $d_i$ )
- Infectious status (not infected – symptomatic – asymptomatic) ( $s_i$ )
- Protection status (yes – no) ( $p_i$ )
- Study period (alpha – omicron)

## ■ Protection

Un individu est considéré protégé si :

- Infections passées : infection de moins de 6 mois et +30j
- Vaccination : Alpha: 1<sup>e</sup> dose +10j avant et -90j OU 2<sup>e</sup> dose +10j avant  
Omicron: 2<sup>e</sup> dose +10j avant

## ■ Tranches d'âge

Bas âge (<6 ans)

Enfance (6-11 ans)

Ados/adultes (12+ ans)

# MODEL

- **Data augmentation:** unobserved infection date ( $\tau_i$ ):

Symptoms onset date / 1<sup>st</sup> test date – incubation period (distribution  $g$ )

## Distribution of incubation period:

Symptomatic : distributions from S. Galmiche et al.

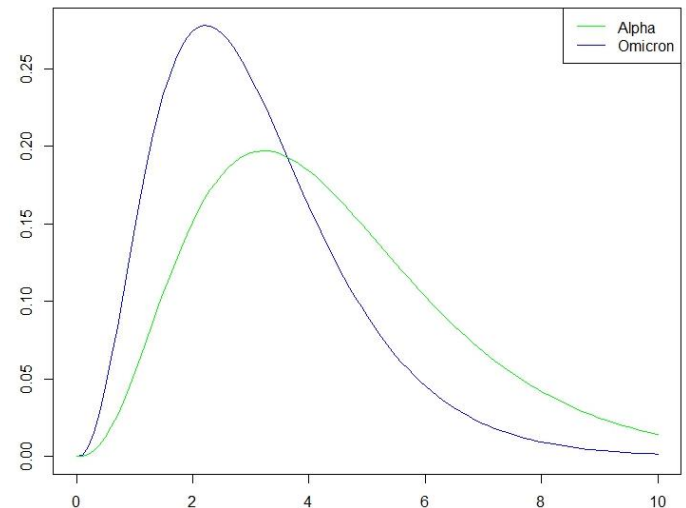
ALPHA : gamma(mean=4.42, sd=2.3)

OMICRON : gamma(mean=3.09, sd=1.64)

Asymptomatic :

ALPHA : Unif(2,7)

OMICRON : Unif(1,5)



# MODEL

## ■ Risk of infection of individual i by j at time t :

$$h_{j \rightarrow i}(t) = \underbrace{\frac{\beta}{\binom{n_i}{4}^\delta}}_{\text{Baseline transmission rate dependent on household size } (\delta)} \times \underbrace{\mu_{susc}(a_i) \times \mu_{acq}(p_i)}_{\text{Relative susceptibility of infected i : By age of i } (\mu_{susc}) \text{ By protection of i } (\mu_{acq})} \times \underbrace{\mu_{inf}(s_j, a_j) \times \mu_{transm}(p_j)}_{\text{Relative infectiousness of infector j : By age and symptomatic status of j } (\mu_{inf}) \text{ By protection of j } (\mu_{transm})} \times \underbrace{f(t - \tau_j)}_{\text{Likelihood of generation time}}$$

Baseline transmission rate  
dependent on household  
size ( $\delta$ )

Relative susceptibility of infected i :  
By age of i ( $\mu_{susc}$ )  
By protection of i ( $\mu_{acq}$ )

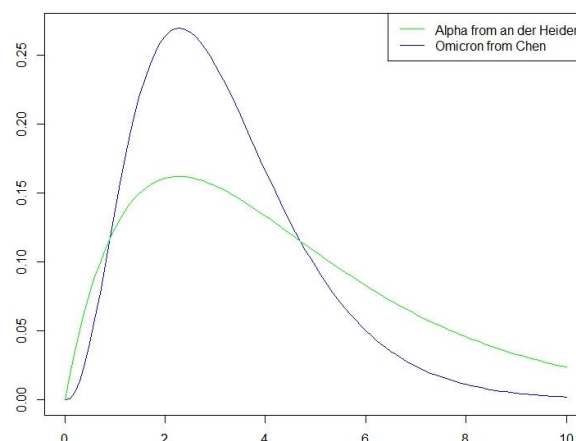
Relative infectiousness of infector j :  
By age and symptomatic status of j ( $\mu_{inf}$ )  
By protection of j ( $\mu_{transm}$ )

Likelihood of  
generation time

### Distribution of generation time:

ALPHA : distribution from an der Heiden et al.  
→  $\text{gamma}(\text{shape}=2, \text{rate}=0.44)$

OMICRON : distribution from Chen et al.  
→  $\text{gamma}(\text{shape}=3.531, \text{rate}=1.098)$



# MODEL

- Estimated parameters:  $\theta = (\beta, \delta, \mu_s^2, \mu_a, \mu_i^5, \mu_t, \alpha)$
- **Likelihood of the data:**

$$\mathcal{L}(\Omega|\theta) = \underbrace{\prod_{i \text{ infected}} g(D_i - \tau_i | s_i)}_{\text{Incubation process (or detection for asymptomatic individuals)}} \underbrace{\prod_{i \text{ secondary case}} \lambda_i(\tau_i) e^{-\int_{\tau_1}^{\tau_i} \lambda_i(u) du}}_{\text{Transmission process}} \underbrace{\prod_{i \text{ not infected}} e^{-\int_{\tau_1}^{t_{\text{end}}} \lambda_i(u) du}}_{\text{Transmission process}}$$

Contribution of infected:  
Survival until time of infection
Contribution non infected:  
Survival until end of household follow-up

# STATISTICAL INFERENCE

Metropolis-Hastings algorithm:

- Update parameters  $\theta$  by  $\theta$ 
  - Proposal for delta (natural scale):  $new = old + \mathcal{N}(0, \sigma^2)$
  - Proposal for others (log scale):  $new = old \cdot e^{\mathcal{N}(0, \sigma^2)}$
- Update augmented data
  - Proposal symptoms/1st test date:  $D_i = d_i + U(0,1)$
  - Proposal infection date:  $\tau_i = D_i - rn(g)$

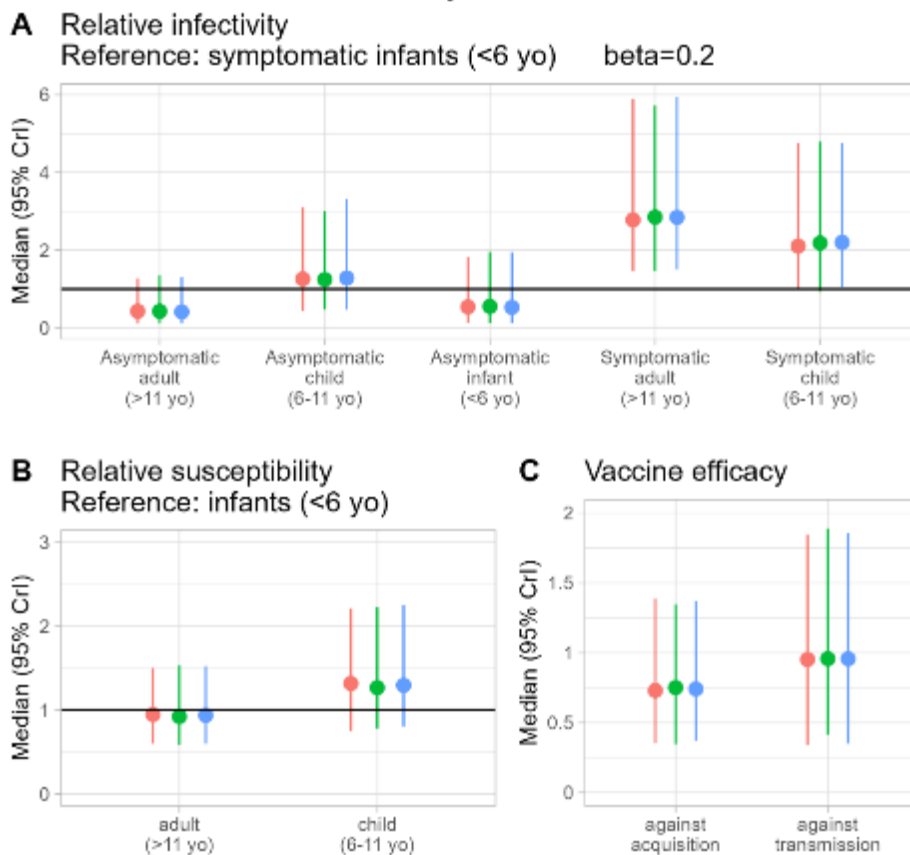
Distributions for priors :

- alpha (risk in community) :  $\text{unif}(0,0.1)$
- delta (dependence hh size) :  $\text{unif}(-3,3)$
- beta (baseline transmission risk) :  $\text{unif}(0,10)$
- $\mu_i, \mu_s$  (relative infectio and susc) :  $\text{logN}(0,1)$
- $\mu_t, \mu_a$  (protection efficacy) :  $\text{logN}(0,1)$

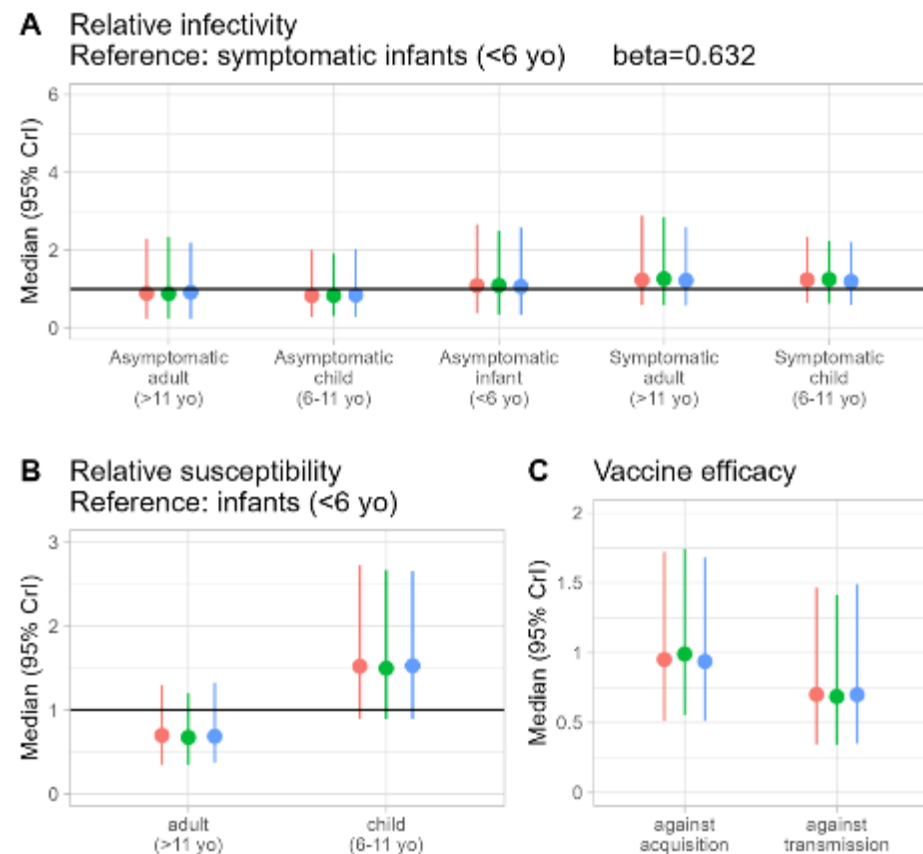
- 500 000 iterations
- Sampling every 500 iterations
- Burn-in of 50 000 iterations
- Evaluation of convergence: visual of chains, probability of acceptance and ess

# PROTECTION EFFICACY

## alpha



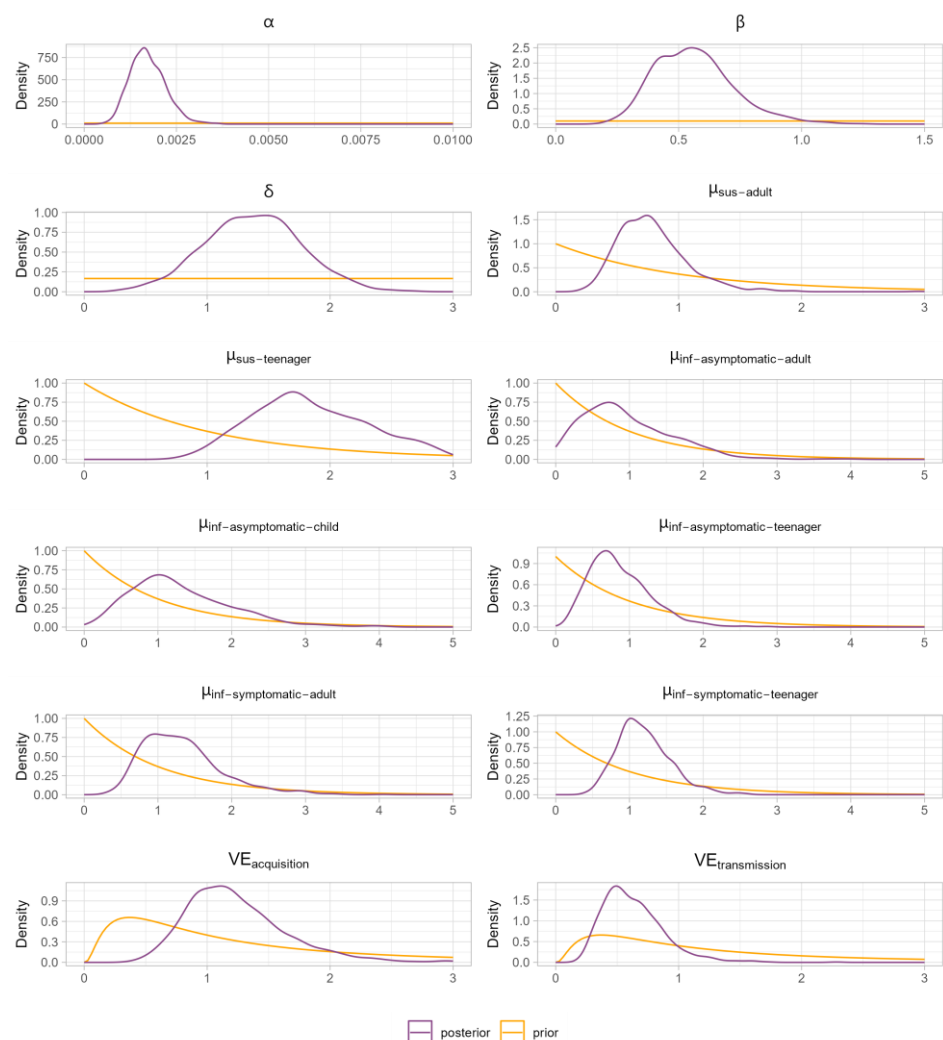
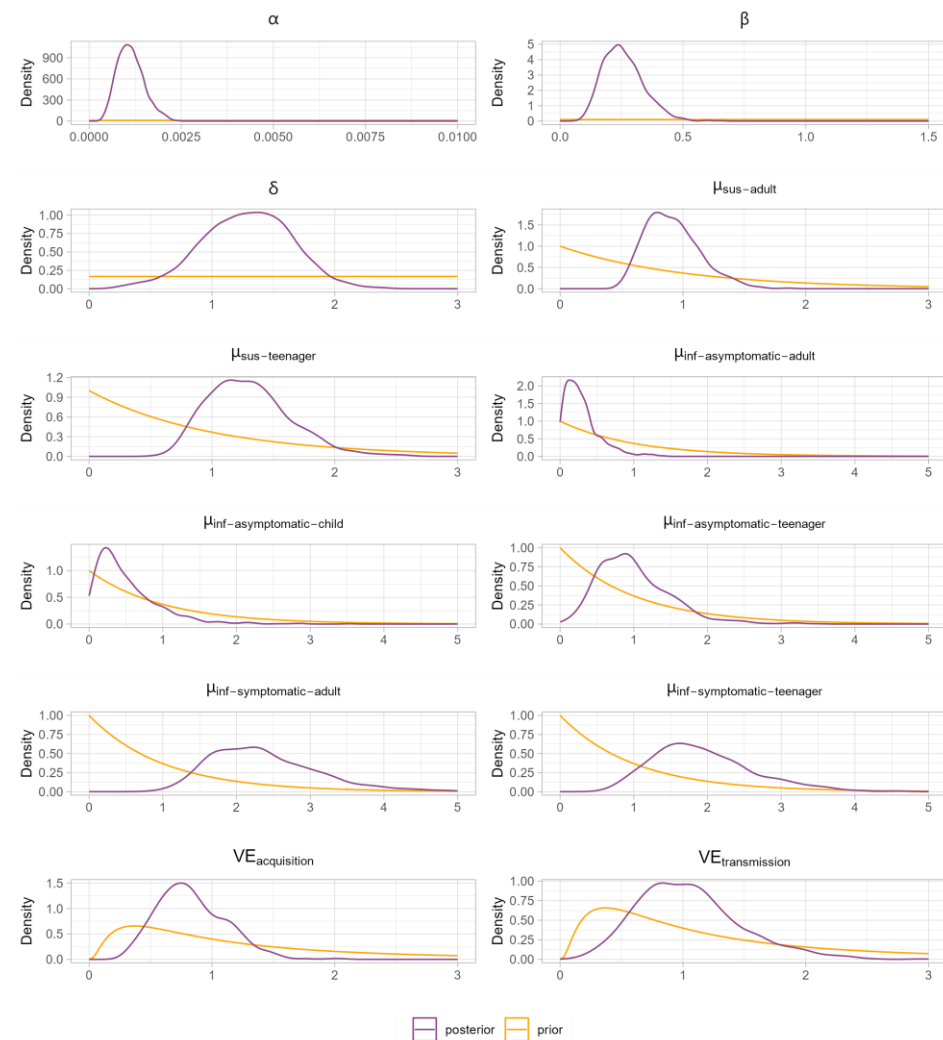
## omicron



# COMPARAISON PRIOR/POSTERIOR

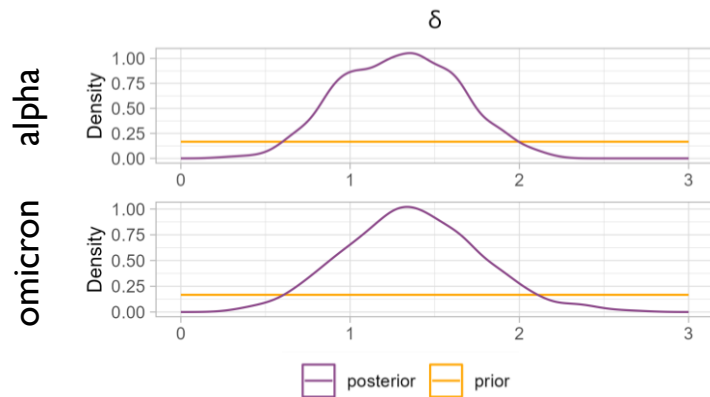
## ALPHA

## OMICRON



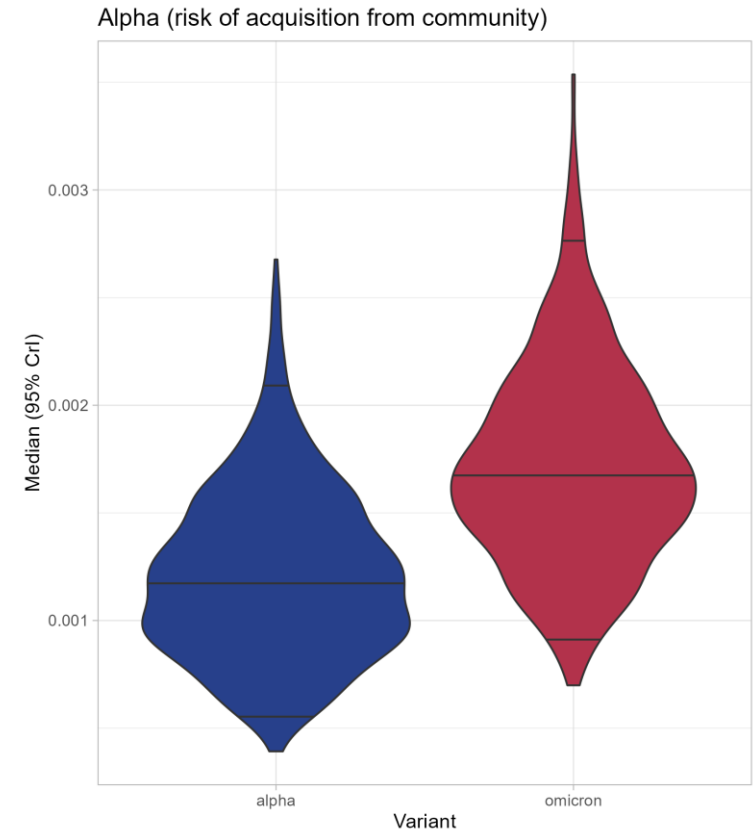
# OTHER ESTIMATED PARAMETERS

- Alpha (risk of acquisition from community)
- Delta (dependance of transmission risk on household size)



DELTA estimates (dependance of beta on household size)

| database | median | q2_5 | q97_5 |
|----------|--------|------|-------|
| alpha    | 1.30   | 0.65 | 1.94  |
| omicon   | 1.36   | 0.63 | 2.20  |







# SIMULATION STUDY

# SIMULATION OF OUTBREAKS

## ■ Choisir un individu index au hasard

- Attribuer une date d'infection à 0
- Déterminer son statut symptomatique (probabilité fixée)
- Déterminer sa date de symptômes (grâce à la distribution d'incubation)

## ■ Pour chaque pas de temps $t$ (pendant la durée de suivi) :

- Pour tous les individus susceptibles  $i$  du foyer, faire :
  - Calculer leur probabilité d'infection  $\lambda_i(t)$  avec les paramètres fixés  $(\alpha, \beta, \delta, \mu_i^5, \mu_s^2, \mu_a, \mu_t)$
  - Déterminer s'il est infecté ou non
    - Attribuer la date d'infection correspondant à l'itération
    - Déterminer le statut symptomatique
    - Déterminer sa date de symptômes

## ■ Choisir un cas d'inclusion au hasard :

- Lui attribuer (ainsi qu'au reste du foyer) une date d'inclusion en fonction du délai symptômes – inclusion des cas d'inclusion des données

# SIMULATION OF OUTBREAK

## ■ Chose randomly an index individual

- Assign an infection date (date\_inf)=0
- Draw  $u \sim \text{Unif}(0,1)$   
If  $u < p_{\text{asympto}}$ , assign asympto status (infect\_status=2)  
Else, infect\_status=1 (sympto)
- Assign symptoms onset date with incubation period distributions

$p_{\text{asympto}}=0.22$  (alpha)  
 $p_{\text{asympto}}=0.25$  (omicron)

*Sympto or asympto?*

## ■ For each time step t (for duration followup iterations), do:

- For all susceptible individuals i in household, do:
  - Compute proba\_inf  $\lambda_i(t)$  with fixed parameters  $(\alpha, \beta, \delta, \mu_i^5, \mu_s^2, \mu_a, \mu_t)$
  - Draw  $u \sim \text{Unif}(0,1)$ . *Infecté ou non ?*  
If  $u < \text{proba\_inf}$ ,
    - Assign date of infection corresponding to the iteration.
    - Draw  $u \sim \text{Unif}(0,1)$ .  
If  $u < p_{\text{asympto}}$ , assign infect\_status=2 (asympto).  
Else, infect\_status=1 (sympto).
  - - Assign symptoms onset date date\_sympt with incubation period distributions  
Else, infect\_status=0.

$$\lambda_i(t) = \alpha + \sum_{\substack{j \in H_i, \\ \tau_j < t}} h_{j \rightarrow i}(t)$$

*Sympto ou asympto ?*

## ■ Chose randomly an inclusion case:

- Assign inclusion date from delay symptoms–inclusion of inclusion cases from Pedcovid

|                            |                              |
|----------------------------|------------------------------|
| <b>Alpha (84 foyers) :</b> | <b>Omicron (46 foyers) :</b> |
| 67 incluCase sympto        | 36 incluCase sympto          |
| 53 incluCase aussi index   | 37 incluCase aussi index     |

?

## Scénarios d'inclusion des foyers

### Recrutement via **enfants** (PedCovid-like)



- Si le cas d'inclusion est un adulte → exclusion
- Si le cas d'inclusion est un enfant :
  - Si symptômes après inclusion → exclusion
  - Sinon → recrutement  
(reflétant les nombres de symptomatiques et cas index des cas d'inclusion de PedCovid)

### Recrutement **aléatoire** (Random)



- Pas de classification/exclusion
- Tirer un nombre total au hasard

### Recrutement via **adultes** (Via adults)



- Si le cas d'inclusion est un **enfant** → exclusion
- Si le cas d'inclusion est un **adulte** :
  - Si symptômes après inclusion → exclusion
  - Sinon → recrutement  
(reflétant les nombres de symptomatiques et cas index des cas d'inclusion de PedCovid)

### **Sur-sampling** avec recrutement via enfants (PedCovid-like + sampling)



- Si le cas d'inclusion est un adulte → exclusion
- **Si aucun adulte infecté dans le foyer → exclusion**
- Si le cas d'inclusion est un enfant :
  - Si symptômes après inclusion → exclusion
  - Sinon → recrutement  
(reflétant les nombres de symptomatiques et cas index des cas d'inclusion de PedCovid)

# Résultats hypothèse nulle

## Période alpha

$\alpha = 0,001$

$\beta = 0,2$

$\delta = 1,3$

$\text{rel\_inf} = 1$

$\text{rel\_susc} = 1$

$\text{VE\_acq} = 0,8$

$\text{VE\_transm} = 1$

## Période omicron

$\alpha = 0,002$

$\beta = 0,5$

$\delta = 1,4$

$\text{rel\_inf} = 1$

$\text{rel\_susc} = 1$

$\text{VE\_acq} = 0,8$

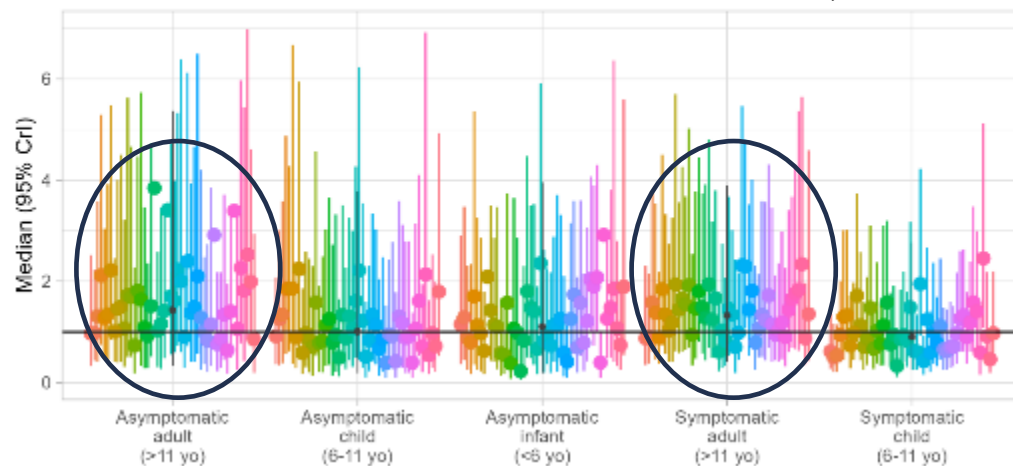
$\text{VE\_transm} = 0,8$

# Résultats hypothèse nulle

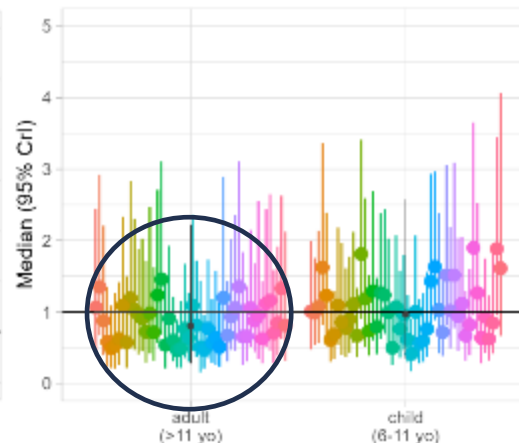
Infectiosité relative = 1  
Susceptibilité relative = 1

## Recrutement PedCovid-like (omicron)

Relative infectivity  
Reference: symptomatic infants (<6 yo)  $\beta = 0.379$  (simu=0.5)



B Relative susceptibility  
Reference: infants (<6 yo)

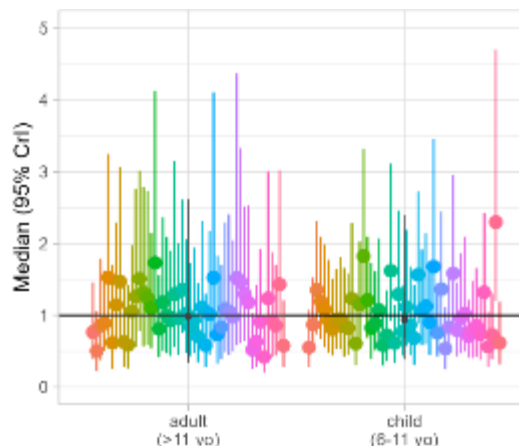
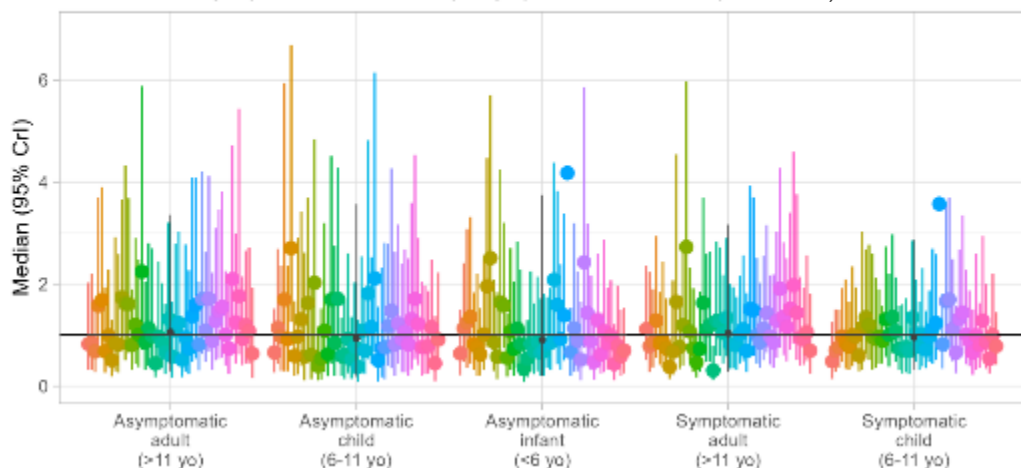


**Recrutement PedCovid-like :**  
Sur-estimation de l'infectiosité des adultes

Sous-estimation de la susceptibilité des adultes

## Recrutement aléatoire (omicron)

Relative infectivity  
Reference: symptomatic infants (<6 yo)  $\beta = 0.497$  (simu=0.5)

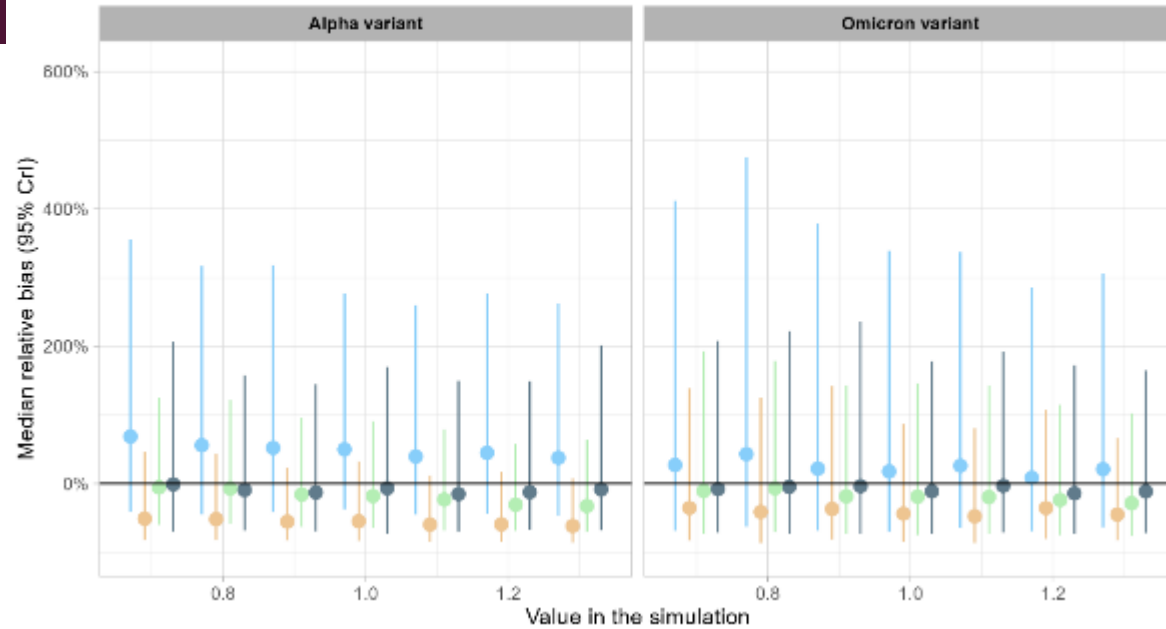


**Recrutement aléatoire :**  
Bonne estimation

## Bias for Infectivity of Symptomatic Adults

50 aggregated results for each simulated value

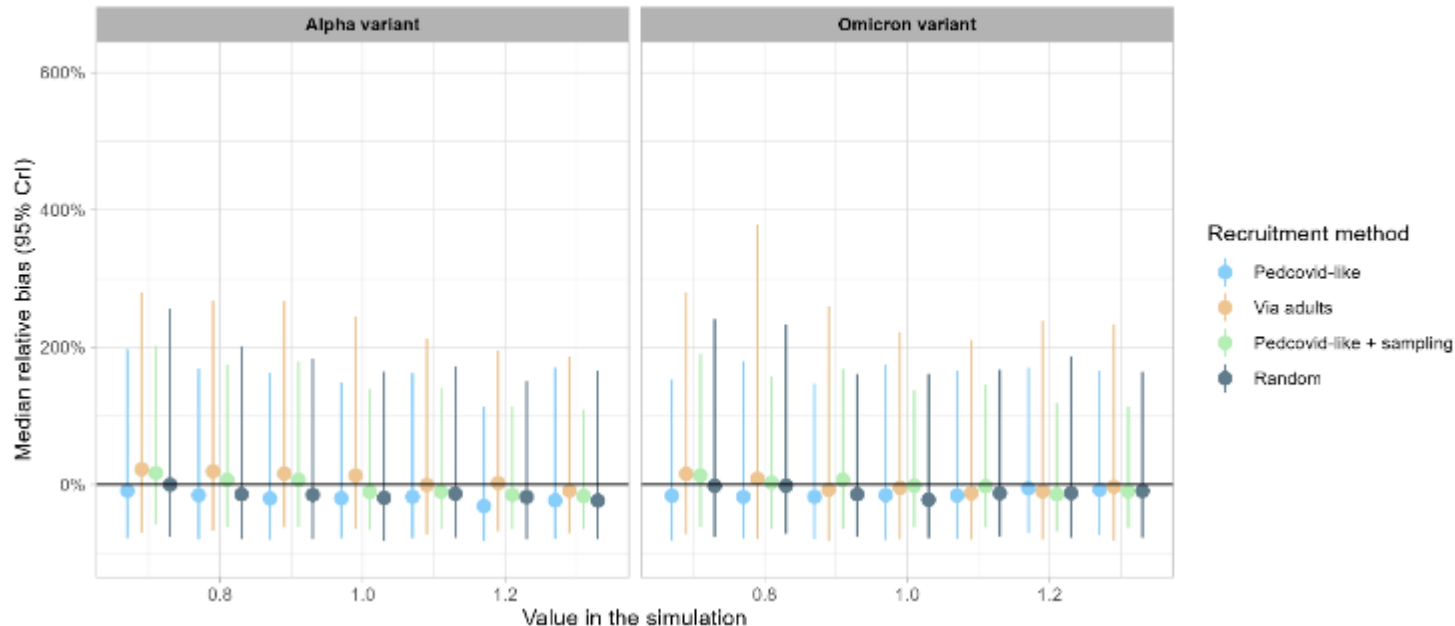
Bias = (estimated value - simulation value) / simulation value



## Bias for Infectivity of Symptomatic Children

50 aggregated results for each simulated value

Bias = (estimated value - simulation value) / simulation value

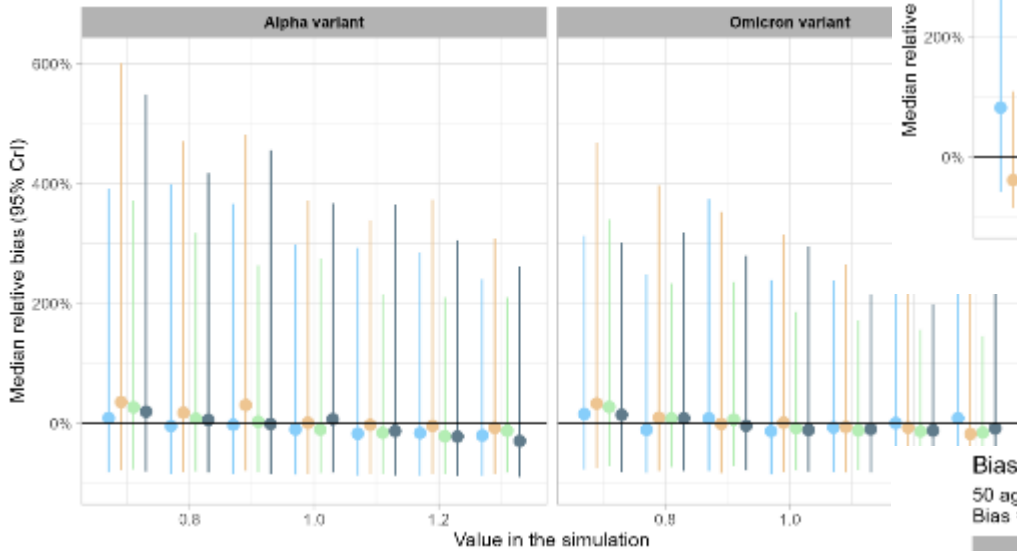




### Bias for Infectivity of Asymptomatic Infants

50 aggregated results for each simulated value

Bias = (estimated value - simulation value) / simulation value



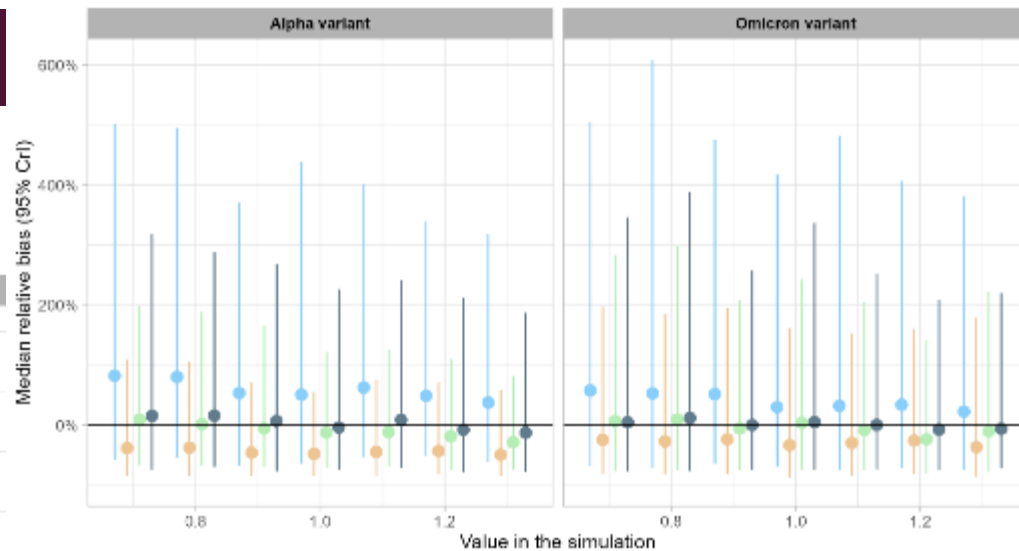
### Recruitment method

- Pedcovid-like
- Via adults
- Pedcovid-like + sampling
- Random

### Bias for Infectivity of Asymptomatic Adults

50 aggregated results for each simulated value

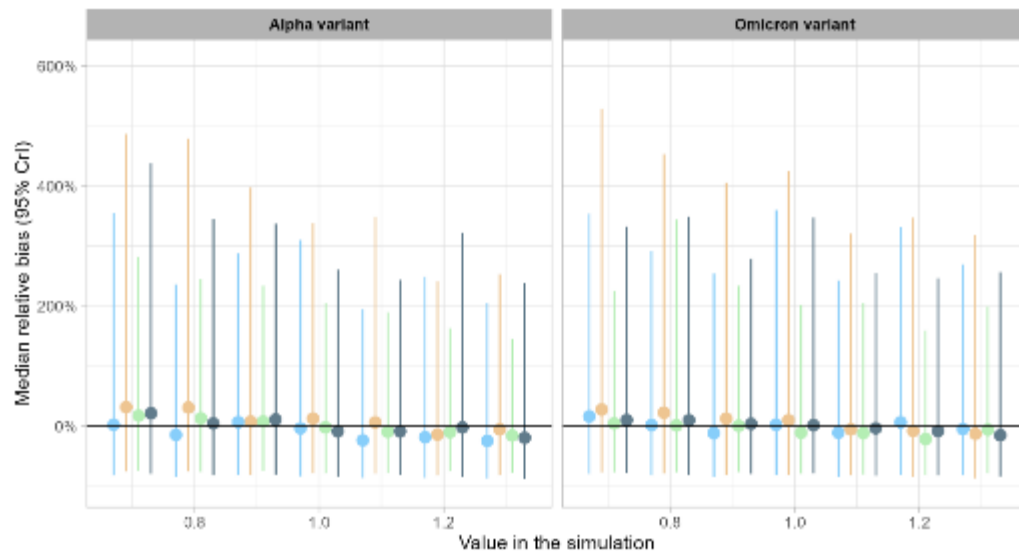
Bias = (estimated value - simulation value) / simulation value



### Bias for Infectivity of Asymptomatic Children

50 aggregated results for each simulated value

Bias = (estimated value - simulation value) / simulation value



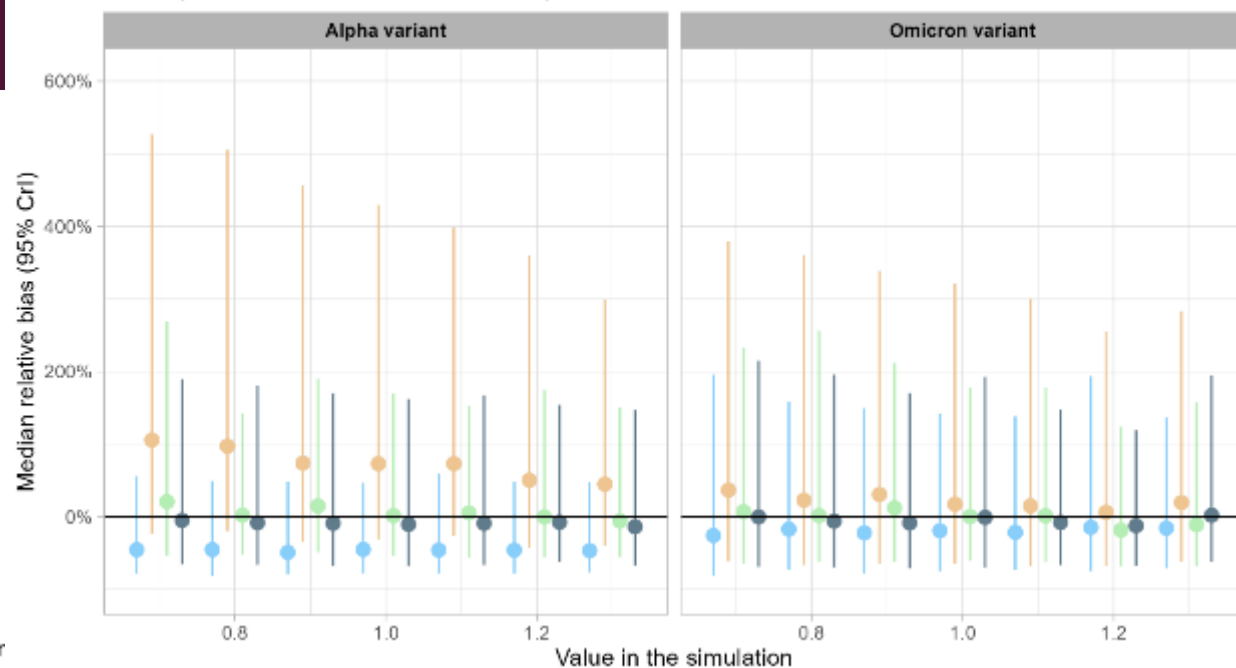




### Bias for Susceptibility of Adults

50 aggregated results for each simulated value

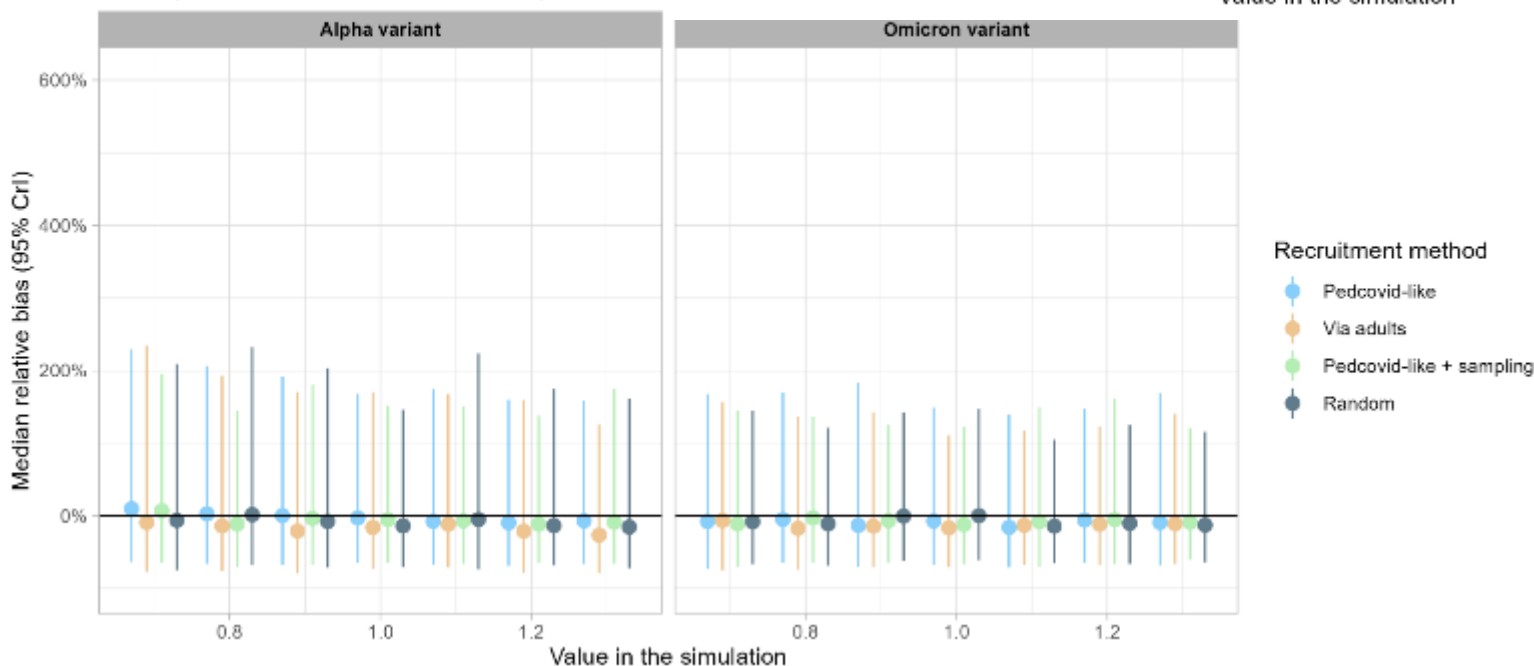
Bias = (estimated value - simulation value) / simulation value



### Bias for Susceptibility of Children

50 aggregated results for each simulated value

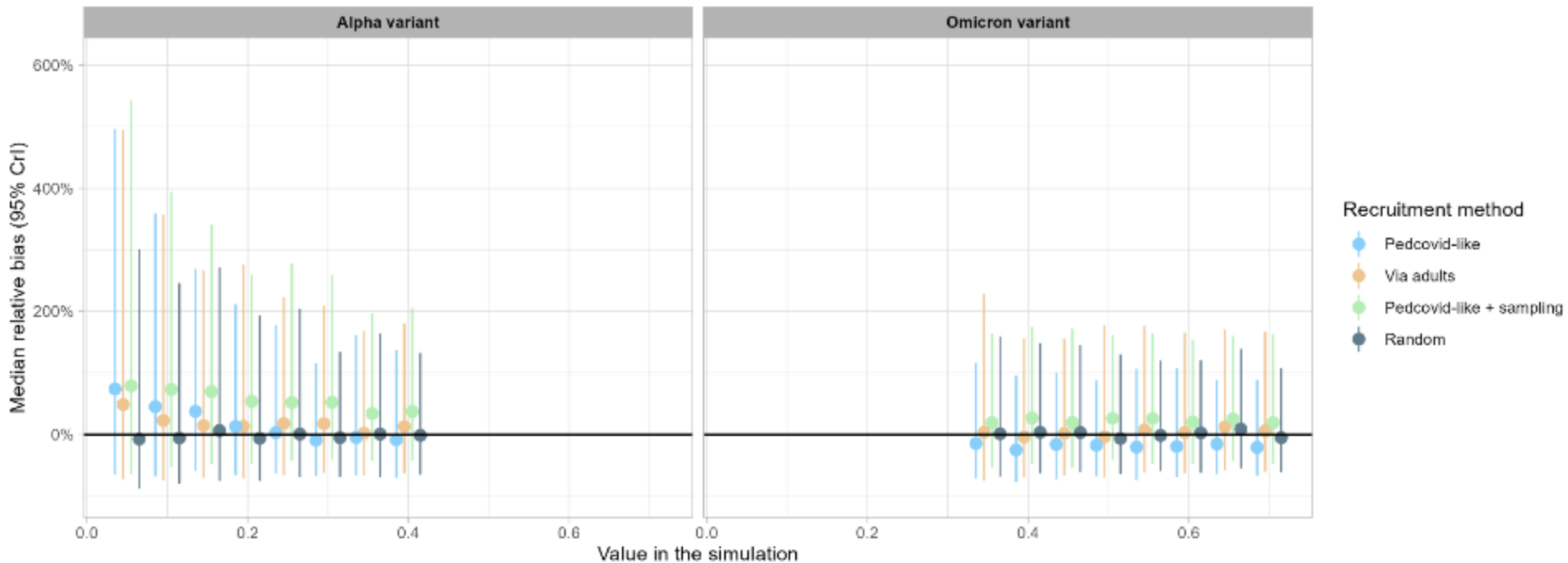
Bias = (estimated value - simulation value) / simulation



## Bias for transmission rate (beta)

50 aggregated results for each simulated value

Bias = (estimated value - simulation value) / simulation value





# SENSIBILITY ANALYSIS ON MU\_INF AND MU\_SUSC PRIORS

# SENSIB ON MU\_INF AND MU\_SUSC

| ALPHA PERIOD           |          |          |                   |         |          |           |          |                 |          |          |
|------------------------|----------|----------|-------------------|---------|----------|-----------|----------|-----------------|----------|----------|
|                        | Exp(1)   |          |                   |         |          | logN(0,1) |          |                 |          |          |
|                        | Mean     | Median   | CrI ETI 95%       | Best    | Mode     | Mean      | Median   | CrI ETI 95%     | Best     | Mode     |
| alpha - prior          |          |          |                   |         |          |           |          |                 |          |          |
| prior                  | 1        | 0.69     | 0.03-3.68         | -       | 0        | 1.65      | 1        | 0.14-7.10       | -        | 0.37     |
| alpha - susceptibility |          |          |                   |         |          |           |          |                 |          |          |
| rSC                    | 1.3      | 1.27     | 0.78-2.07         | 0.56    | 1.15     | 1.27      | 1.23     | 0.76-2.01       | 1.74     | 1.15     |
| rSA                    | 0.92     | 0.89     | 0.57-1.41         | 0.65    | 0.79     | 0.9       | 0.87     | 0.56-1.35       | 1.03     | 0.84     |
| alpha - infectiousness |          |          |                   |         |          |           |          |                 |          |          |
| rInfAA                 | 0.29     | 0.24     | 0.01-0.92         | 0.55    | 0.13     | 0.42      | 0.35     | 0.1-1.13        | 0.17     | 0.24     |
| rInfAC                 | 1        | 0.92     | 0.28-2.21         | 0.54    | 0.89     | 1.25      | 1.13     | 0.36-3          | 0.88     | 0.9      |
| rInfAI                 | 0.51     | 0.38     | 0.03-1.62         | 0.88    | 0.22     | 0.65      | 0.55     | 0.12-1.78       | 0.5      | 0.41     |
| rInfSA                 | 2.38     | 2.28     | 1.29-4.12         | 2.37    | 2.22     | 2.93      | 2.71     | 1.41-5.57       | 2.66     | 2.3      |
| rInfSC                 | 1.94     | 1.85     | 0.9-3.56          | 2.04    | 1.62     | 2.41      | 2.19     | 1-5.26          | 1.89     | 1.99     |
| alpha - logLikelihood  |          |          |                   |         |          |           |          |                 |          |          |
| logLik                 | -1404.96 | -1404.57 | -1421.34--1390.12 | -1434.3 | -1403.64 | -1404.52  | -1404.54 | -1421.8--1388.9 | -1430.11 | -1405.91 |

ETI = equal-tailed interval (quantiles 2,5% and 97,5%) → 95% of points fall in this interval

Best = param that gives highest likelihood

Mode = max of the density curve

# SENSIB ON MU\_INF AND MU\_SUSC

| OMICRON PERIOD           |         |         |                 |         |         |           |         |                 |        |         |
|--------------------------|---------|---------|-----------------|---------|---------|-----------|---------|-----------------|--------|---------|
|                          | Exp(1)  |         |                 |         |         | logN(0,1) |         |                 |        |         |
|                          | Mean    | Median  | CrI ETI 95%     | Best    | Mode    | Mean      | Median  | CrI ETI 95%     | Best   | Mode    |
| omicron - prior          |         |         |                 |         |         |           |         |                 |        |         |
| prior                    | 1       | 0.69    | 0.03-3.68       | -       | 0       | 1.65      | 1       | 0.14-7.10       | -      | 0.37    |
| omicron - susceptibility |         |         |                 |         |         |           |         |                 |        |         |
| rSC                      | 1.89    | 1.8     | 1.04-3.17       | 0.9     | 1.69    | 1.96      | 1.86    | 1.09-3.39       | 1.11   | 1.73    |
| rSA                      | 0.78    | 0.74    | 0.36-1.45       | 0.8     | 0.74    | 0.8       | 0.77    | 0.39-1.46       | 0.43   | 0.66    |
| omicron - infectiousness |         |         |                 |         |         |           |         |                 |        |         |
| rInfAA                   | 0.96    | 0.84    | 0.11-2.26       | 1.81    | 0.72    | 1.07      | 0.94    | 0.2-2.77        | 1.13   | 0.76    |
| rInfAC                   | 0.87    | 0.79    | 0.26-1.88       | 1.6     | 0.68    | 0.91      | 0.83    | 0.29-1.98       | 0.42   | 0.68    |
| rInfAI                   | 1.31    | 1.19    | 0.3-2.97        | 2.24    | 1.01    | 1.45      | 1.34    | 0.38-3.19       | 1.57   | 1.26    |
| rInfSA                   | 1.33    | 1.25    | 0.57-2.7        | 0.75    | 0.96    | 1.38      | 1.28    | 0.56-2.67       | 1.25   | 1.16    |
| rInfSC                   | 1.2     | 1.15    | 0.61-2.05       | 0.65    | 1.02    | 1.23      | 1.18    | 0.61-2.14       | 1.37   | 1.13    |
| omicron - logLikelihood  |         |         |                 |         |         |           |         |                 |        |         |
| logLik                   | -730.16 | -729.84 | -744.02--717.53 | -753.54 | -729.39 | -729.86   | -729.68 | -743.85--718.14 | -752.6 | -730.05 |

ETI = equal-tailed interval (quantiles 2,5% and 97,5%) → 95% of points fall in this interval

Best = param that gives highest likelihood

Mode = max of the density curve



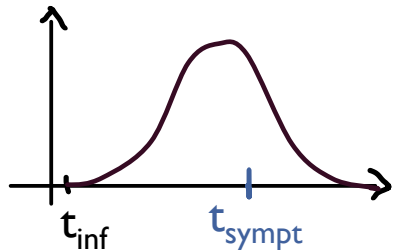
# SENSIBILITY ANALYSIS ON GENERATION TIMES

# SENSIB ON GENERATION TIMES

Baseline model:

- Prior as is:  
 $\alpha$ :  $\text{unif}(0, 0.1)$   
 $\delta$ :  $\text{unif}(-3, 3)$   
 $\beta$ :  $\text{unif}(0, 10)$   
 $\mu_{\text{protect}}$ :  $\log N(0, 1)$   
 $\mu_{\text{inf}}$ :  $\log N(0, 1)$   
 $\mu_{\text{susc}}$ :  $\log N(0, 1)$
- **Generation time:**  
 an der Heiden + Chen  
 without truncation

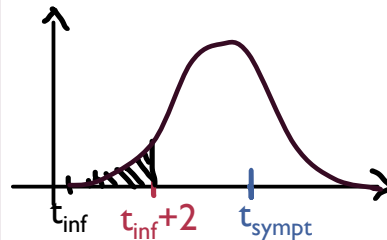
→ No truncation



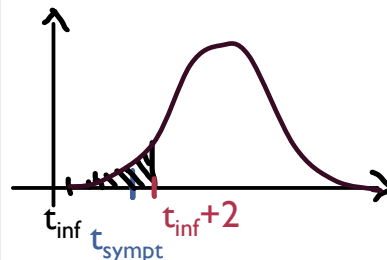
Option 1

→ No infectivity for 1<sup>st</sup> 2 days for alpha and 1<sup>st</sup> day for omicron (not dependent on incubation time)

Cas incub > 2



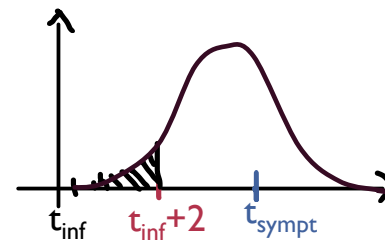
Cas incub < 2



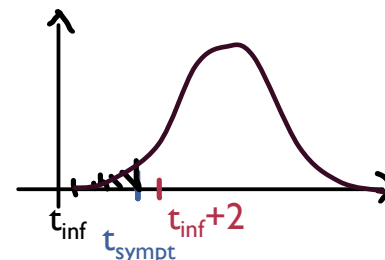
Option 2

→ No infectivity for 1<sup>st</sup> 2 days for alpha and 1<sup>st</sup> day for omicron or until beginning of symptoms if incubation is small

Cas incub > 2

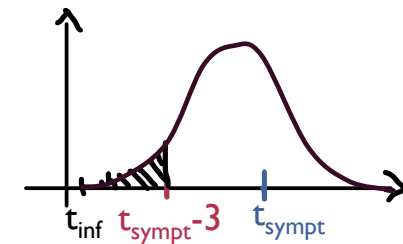


Cas incub < 2



Option 3

→ No infectivity before 3 days before symptoms for alpha and before 2 days before symptoms for omicron (or from infection if incub is small)



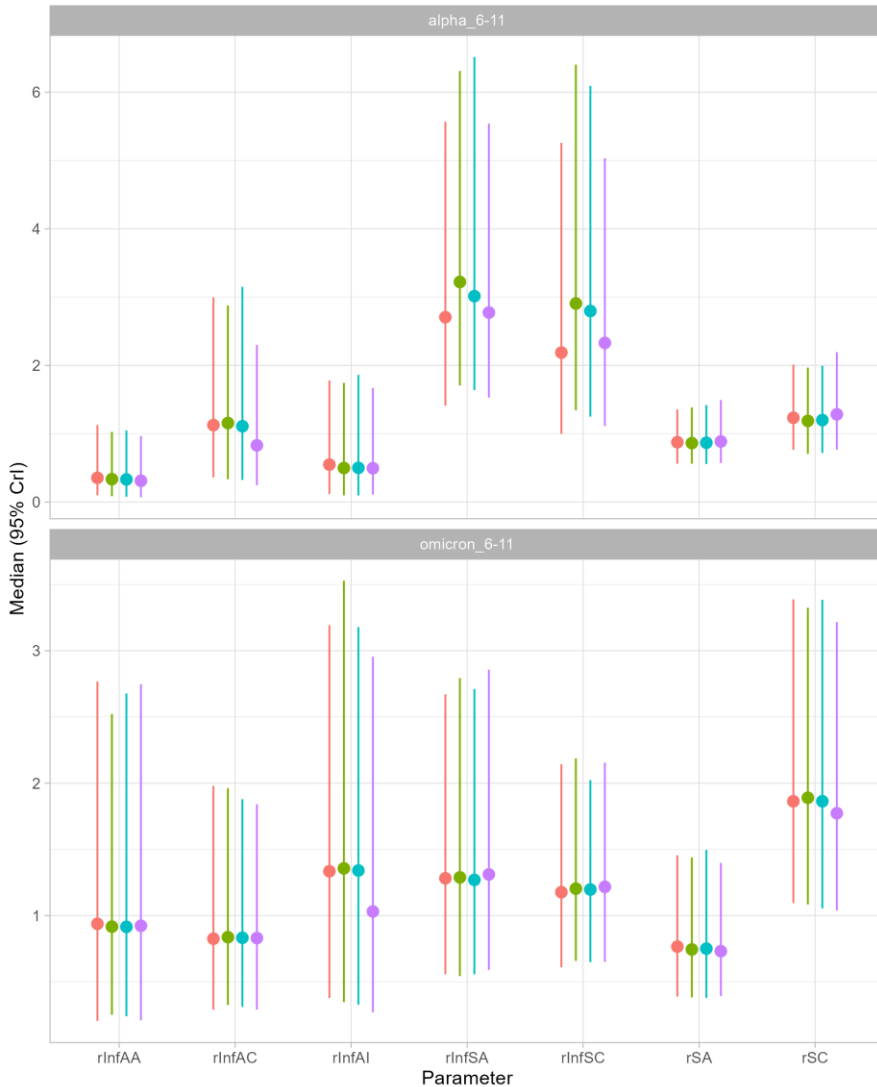
# SENSIB ON GENERATION TIMES

| Characteristic                          | Option 0, N = 901 <sup>†</sup> | Option 1, N = 901 <sup>†</sup> | Option 2, N = 901 <sup>†</sup> | Option 3, N = 901 <sup>†</sup> |
|-----------------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|
| <b>ALPHA period</b>                     |                                |                                |                                |                                |
| logLik                                  | -1,405 (-1,422, -1,389)        | -1,376 (-1,399, -1,358)        | -1,379 (-1,400, -1,360)        | -1,398 (-1,417, -1,382)        |
| rSC                                     | 1.23 (0.76, 2.01)              | 1.19 (0.71, 1.97)              | 1.20 (0.72, 1.99)              | 1.28 (0.76, 2.19)              |
| rSA                                     | 0.87 (0.56, 1.35)              | 0.86 (0.56, 1.38)              | 0.87 (0.56, 1.42)              | 0.89 (0.57, 1.49)              |
| rlnfAA                                  | 0.35 (0.10, 1.13)              | 0.33 (0.08, 1.03)              | 0.33 (0.08, 1.05)              | 0.31 (0.07, 0.97)              |
| rlnfAC                                  | 1.13 (0.36, 3.00)              | 1.16 (0.33, 2.88)              | 1.11 (0.32, 3.15)              | 0.83 (0.24, 2.30)              |
| rlnfAI                                  | 0.55 (0.12, 1.78)              | 0.50 (0.10, 1.74)              | 0.50 (0.10, 1.86)              | 0.49 (0.11, 1.67)              |
| rlnfSA                                  | 2.71 (1.41, 5.57)              | 3.22 (1.71, 6.31)              | 3.01 (1.64, 6.52)              | 2.77 (1.53, 5.54)              |
| rlnfSC                                  | 2.19 (1.00, 5.26)              | 2.91 (1.34, 6.40)              | 2.80 (1.25, 6.09)              | 2.33 (1.11, 5.03)              |
| <b>OMICRON period</b>                   |                                |                                |                                |                                |
| logLik                                  | -730 (-744, -718)              | -721 (-735, -708)              | -721 (-737, -709)              | -727 (-742, -715)              |
| rSC                                     | 1.86 (1.09, 3.39)              | 1.89 (1.08, 3.32)              | 1.86 (1.05, 3.38)              | 1.77 (1.04, 3.22)              |
| rSA                                     | 0.77 (0.39, 1.46)              | 0.74 (0.38, 1.44)              | 0.75 (0.38, 1.50)              | 0.73 (0.39, 1.40)              |
| rlnfAA                                  | 0.94 (0.20, 2.77)              | 0.92 (0.25, 2.52)              | 0.92 (0.24, 2.68)              | 0.92 (0.21, 2.75)              |
| rlnfAC                                  | 0.83 (0.29, 1.98)              | 0.84 (0.32, 1.96)              | 0.83 (0.31, 1.88)              | 0.83 (0.29, 1.84)              |
| rlnfAI                                  | 1.34 (0.38, 3.19)              | 1.36 (0.35, 3.53)              | 1.34 (0.33, 3.18)              | 1.03 (0.27, 2.95)              |
| rlnfSA                                  | 1.28 (0.56, 2.67)              | 1.29 (0.54, 2.79)              | 1.27 (0.56, 2.71)              | 1.31 (0.59, 2.86)              |
| rlnfSC                                  | 1.18 (0.61, 2.14)              | 1.20 (0.66, 2.19)              | 1.20 (0.65, 2.02)              | 1.22 (0.65, 2.15)              |
| <sup>†</sup> Median (eti_low, eti_high) |                                |                                |                                |                                |

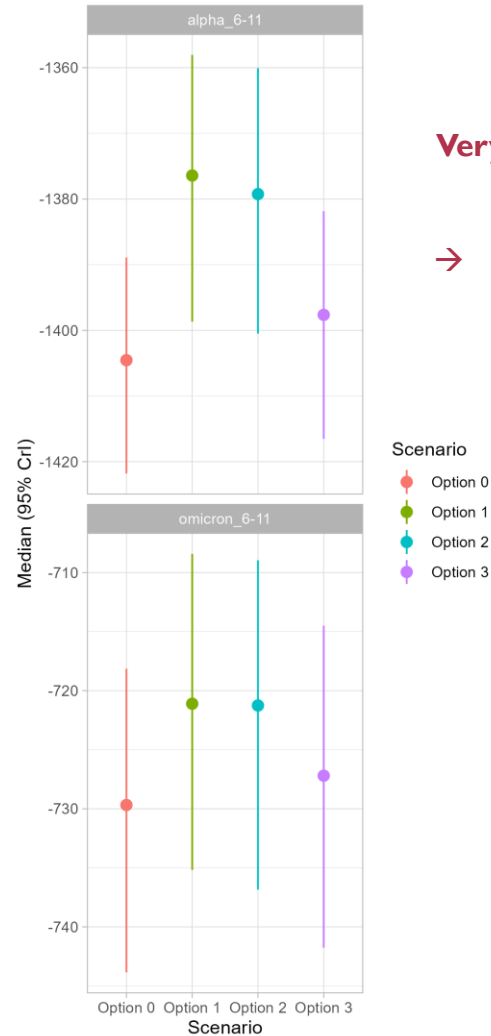


# SENSIB ON GENERATION TIMES

Parameters estimations for 4 scenarios of generation time modelling



LogLikelihoods for 4 scenarios of generation time modelling



**Very similar values for each option, loglikelihood also similar.**

→ **We choose the one with lowest loglikelihood, and the simplest**  
→ **Option 0 (baseline, no truncation)**